

2024-2

Machine Learning COMPETITION

은행 정기 예금 가입 여부 판별하기 ➡



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Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

Lv.6

Lv.7

Lv.8

Team Introduce

| 머신러닝은 이 일(21)을 기억할 것입니다...

허지원

김태현



전처리 및 시각화
도메인 지식 탐색

이용찬



피처 생성
PPT 제작

허지원



코드 정리
모델링



Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

Lv.6

Lv.7

Lv.8

Data Preprocessing

| 데이터 확인 및 전처리

Train data / Test data

모두 결측치 없음

`df_trn.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 31647 entries, 0 to 31646
Data columns (total 18 columns):
#   Column      Non-Null Count  Dtype
---  ---
0    ID          31647 non-null  object
1    age         31647 non-null  int64
2    job         31647 non-null  object
3    marital     31647 non-null  object
4    education   31647 non-null  object
5    default     31647 non-null  object
6    balance     31647 non-null  int64
7    housing     31647 non-null  object
8    loan        31647 non-null  object
9    contact     31647 non-null  object
10   day         31647 non-null  int64
11   month       31647 non-null  object
12   duration    31647 non-null  int64
13   campaign    31647 non-null  int64
14   pdays       31647 non-null  int64
15   previous    31647 non-null  int64
16   poutcome    31647 non-null  object
17   label       31647 non-null  int64
dtypes: int64(8), object(10)
memory usage: 4.3+ MB
```

`df_tst.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 13564 entries, 0 to 13563
Data columns (total 18 columns):
#   Column      Non-Null Count  Dtype
---  ---
0    ID          13564 non-null  object
1    age         13564 non-null  int64
2    job         13564 non-null  object
3    marital     13564 non-null  object
4    education   13564 non-null  object
5    default     13564 non-null  object
6    balance     13564 non-null  int64
7    housing     13564 non-null  object
8    loan        13564 non-null  object
9    contact     13564 non-null  object
10   day         13564 non-null  int64
11   month       13564 non-null  object
12   duration    13564 non-null  int64
13   campaign    13564 non-null  int64
14   pdays       13564 non-null  int64
15   previous    13564 non-null  int64
16   poutcome    13564 non-null  object
17   label       0 non-null      float64
dtypes: float64(1), int64(7), object(10)
memory usage: 1.9+ MB
```



Lv.1

Lv.2

Lv.3

Lv.4

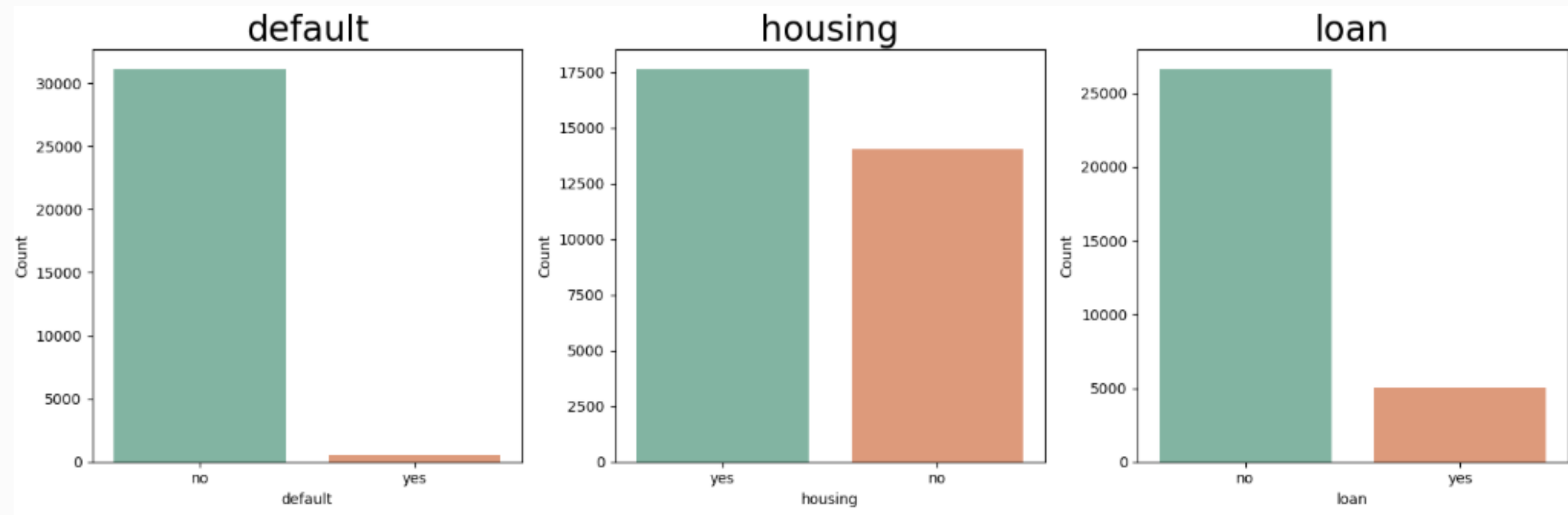
Lv.5

Lv.6

Lv.7

Lv.8

Data Preprocessing | 데이터 확인 및 전처리



Binary Features (Yes / No)



Lv.1

Lv.2

Lv.3

Lv.4

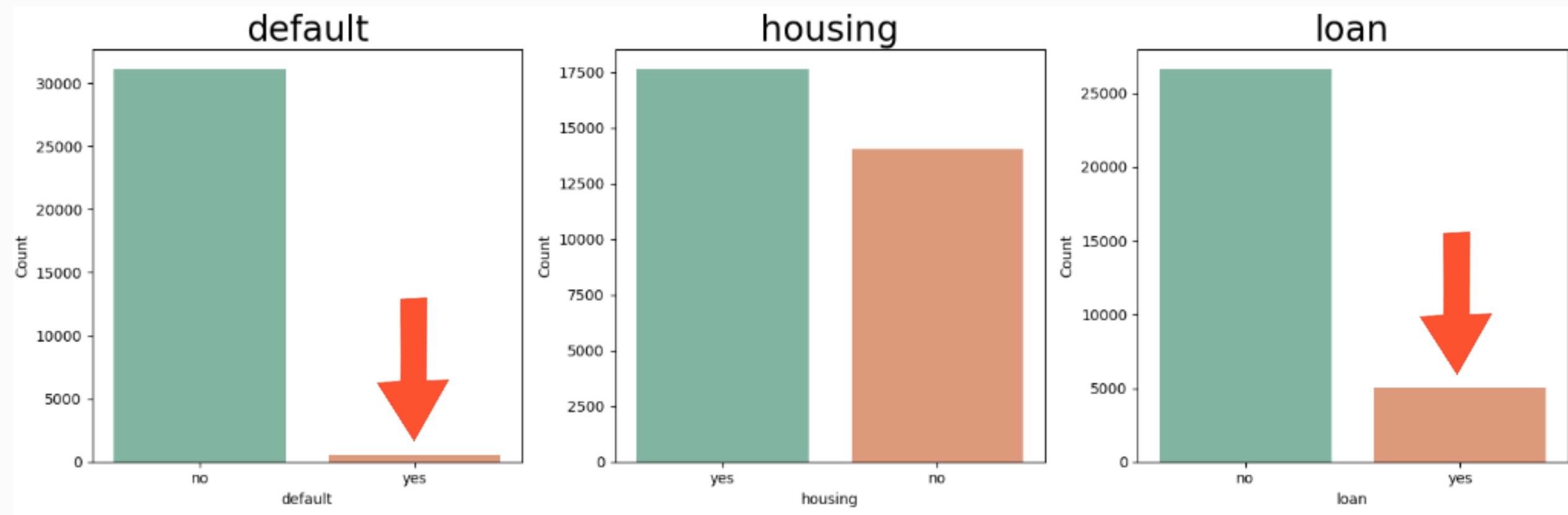
Lv.5

Lv.6

Lv.7

Lv.8

Data Preprocessing | 데이터 확인 및 전처리



Binary Features (Yes / No)



Lv.1

Lv.2

Lv.3

Lv.4

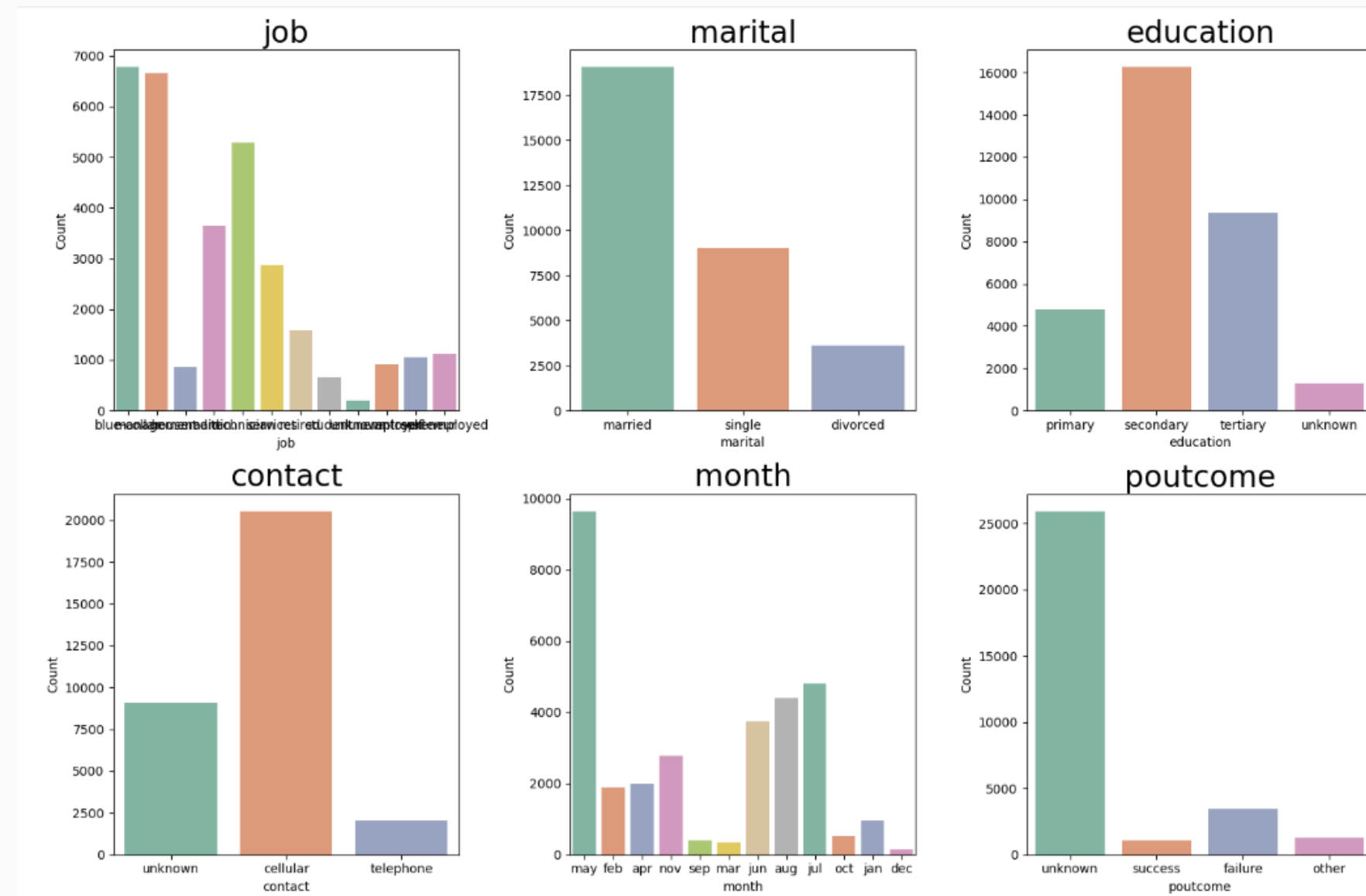
Lv.5

Lv.6

Lv.7

Lv.8

Data Preprocessing | 데이터 확인 및 전처리



Categorical Features



Lv.1

Lv.2

Lv.3

Lv.4

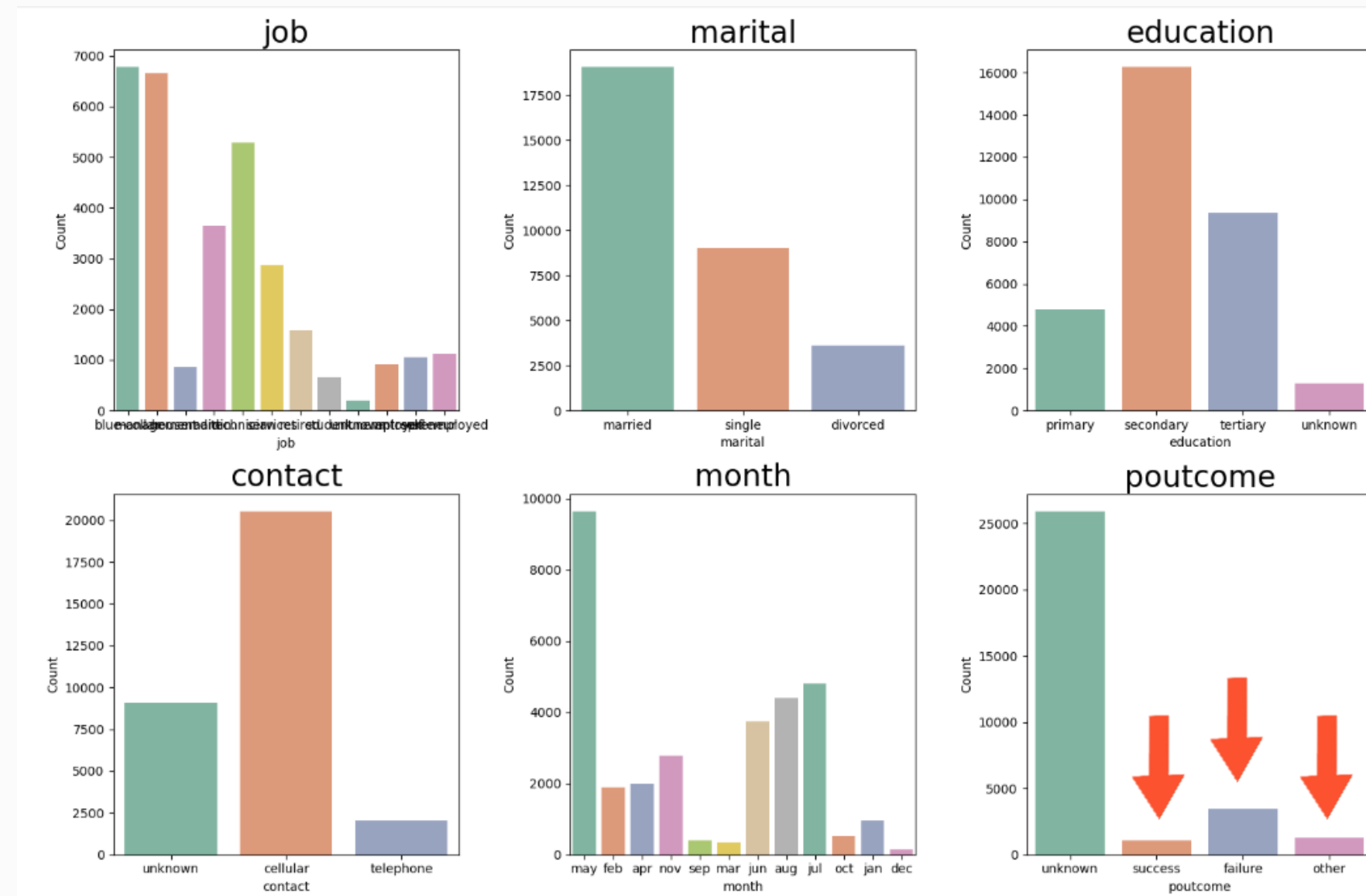
Lv.5

Lv.6

Lv.7

Lv.8

Data Preprocessing | 데이터 확인 및 전처리



Categorical Features



Lv.1

Lv.2

Lv.3

Lv.4

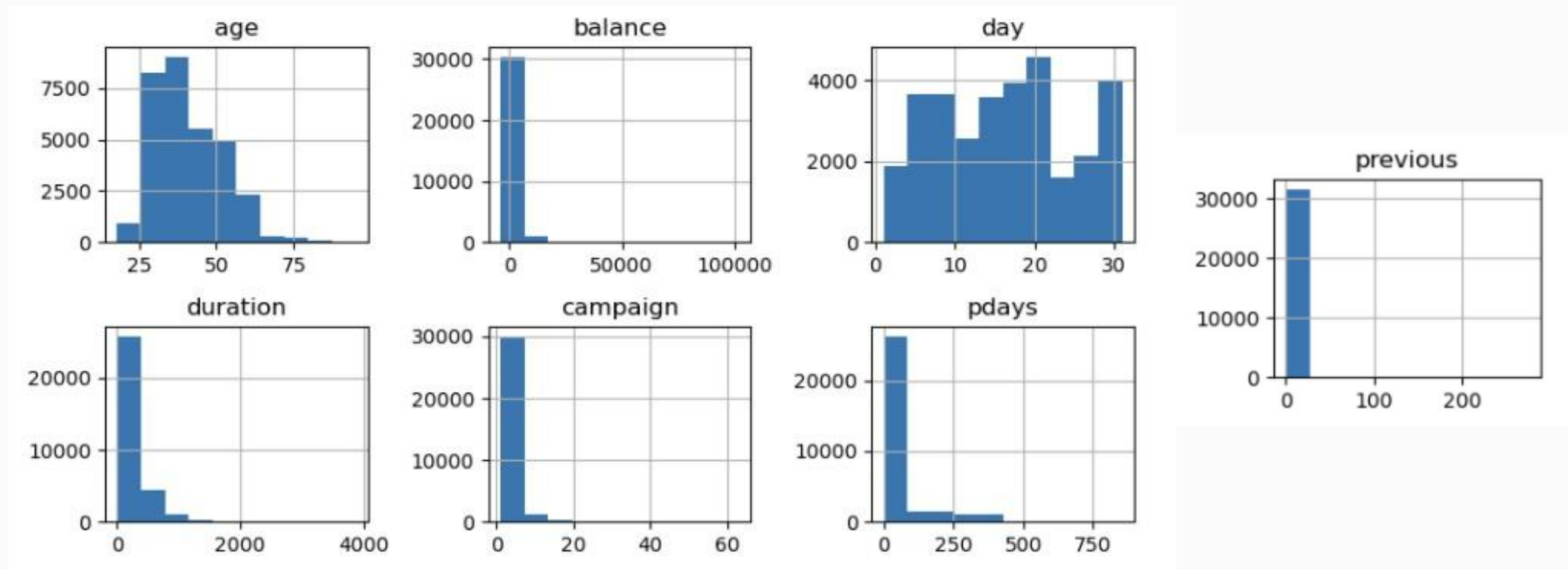
Lv.5

Lv.6

Lv.7

Lv.8

Data Preprocessing | 데이터 확인 및 전처리



Numerical Features



Lv.1

Lv.2

Lv.3

Lv.4

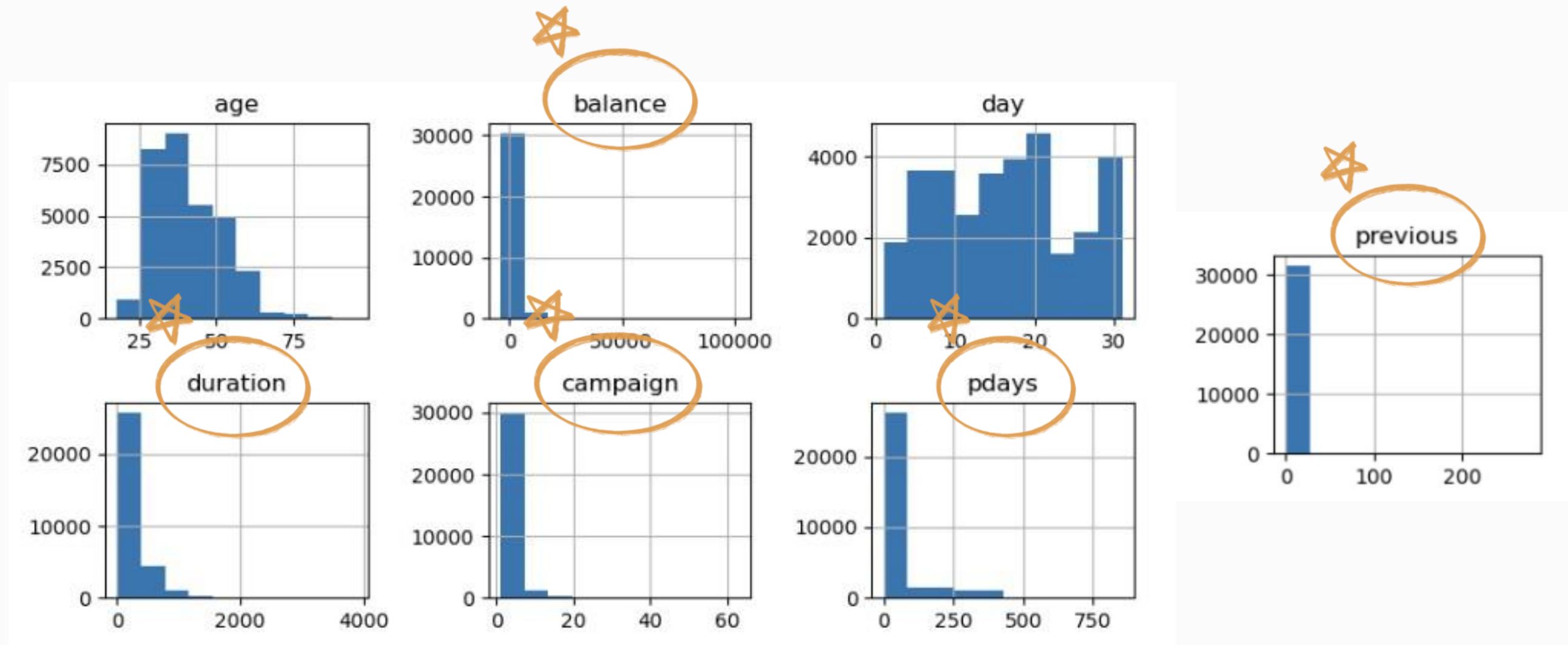
Lv.5

Lv.6

Lv.7

Lv.8

Data Preprocessing | 데이터 확인 및 전처리



Numerical Features



Lv.1

Lv.2

Lv.3

Lv.4

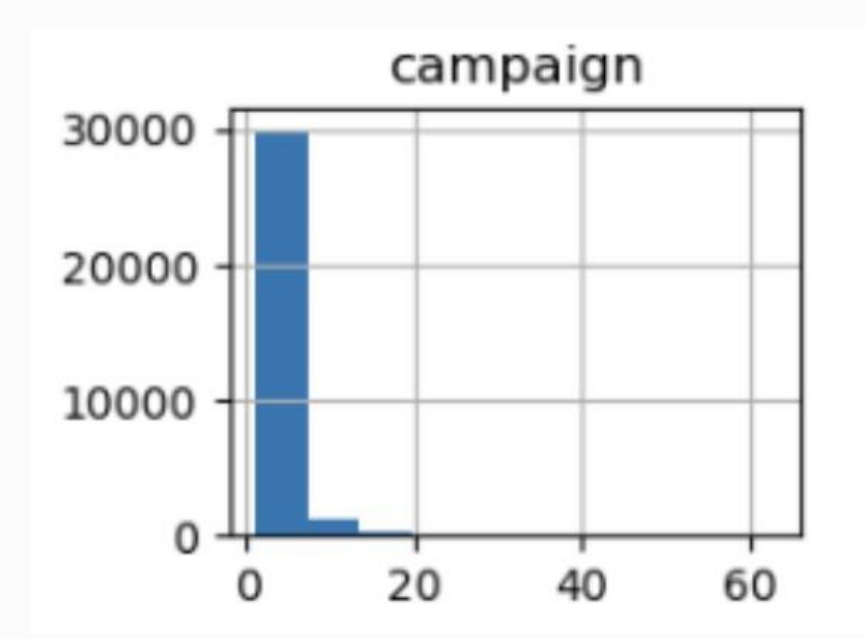
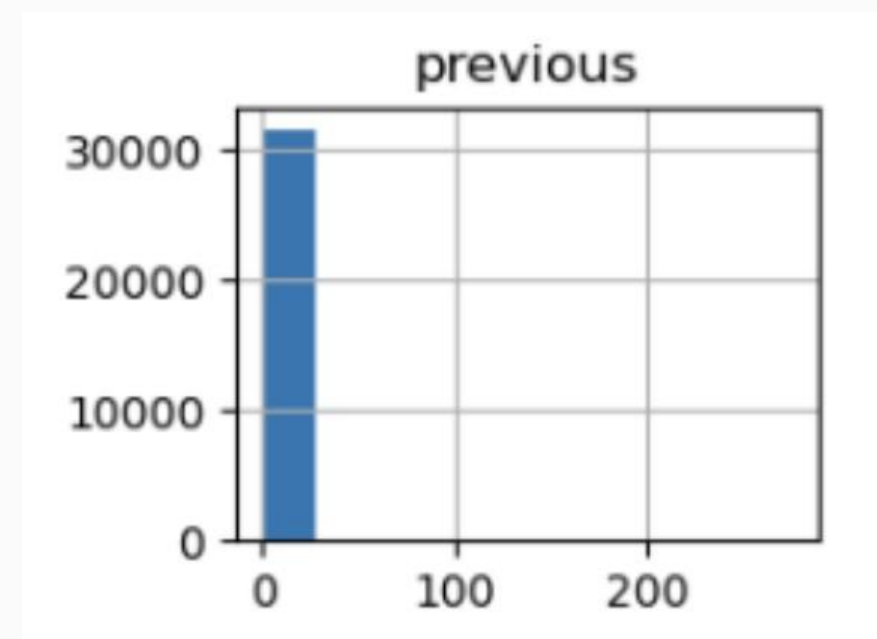
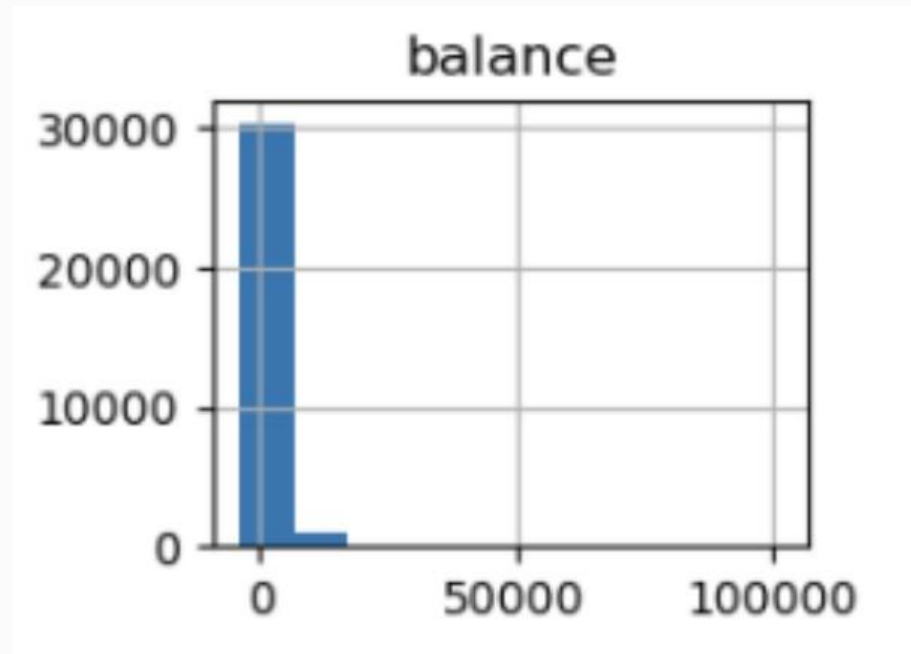
Lv.5

Lv.6

Lv.7

Lv.8

Data Preprocessing | 데이터 확인 및 전처리



위 3개의 피쳐는 **log 변환 적용**

이진 분류?

이진 분류?

➔ 트리 기반 모델

XGBOOST + LightGBM

VOTING

XGBOOST + LightGBM
VOTING



Public 최고 성능

0.65863

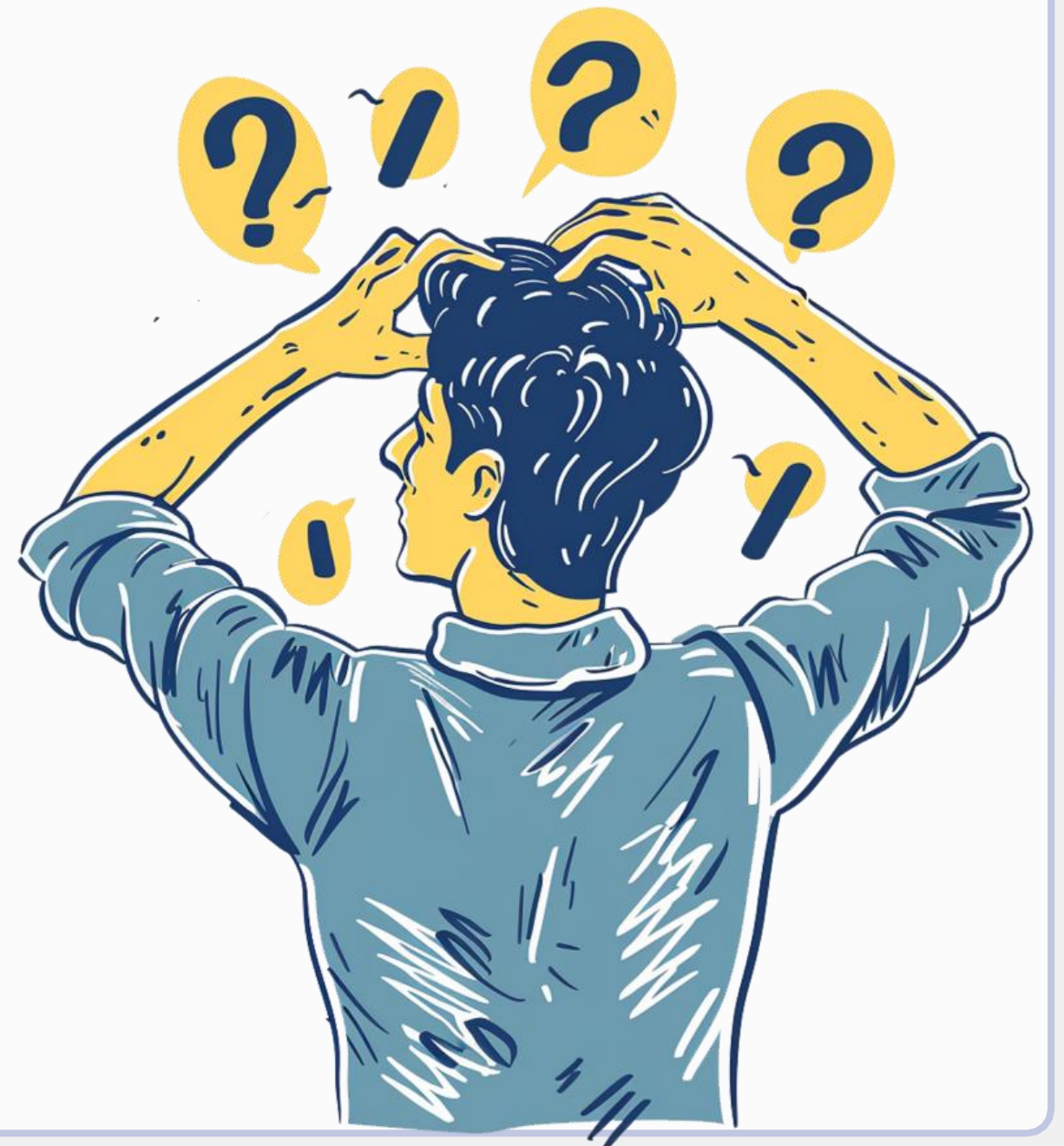
피쳐 제거

하이퍼 파라미터 튜닝

피쳐 생성

학습 모델 변경

데이터 불균형 해소



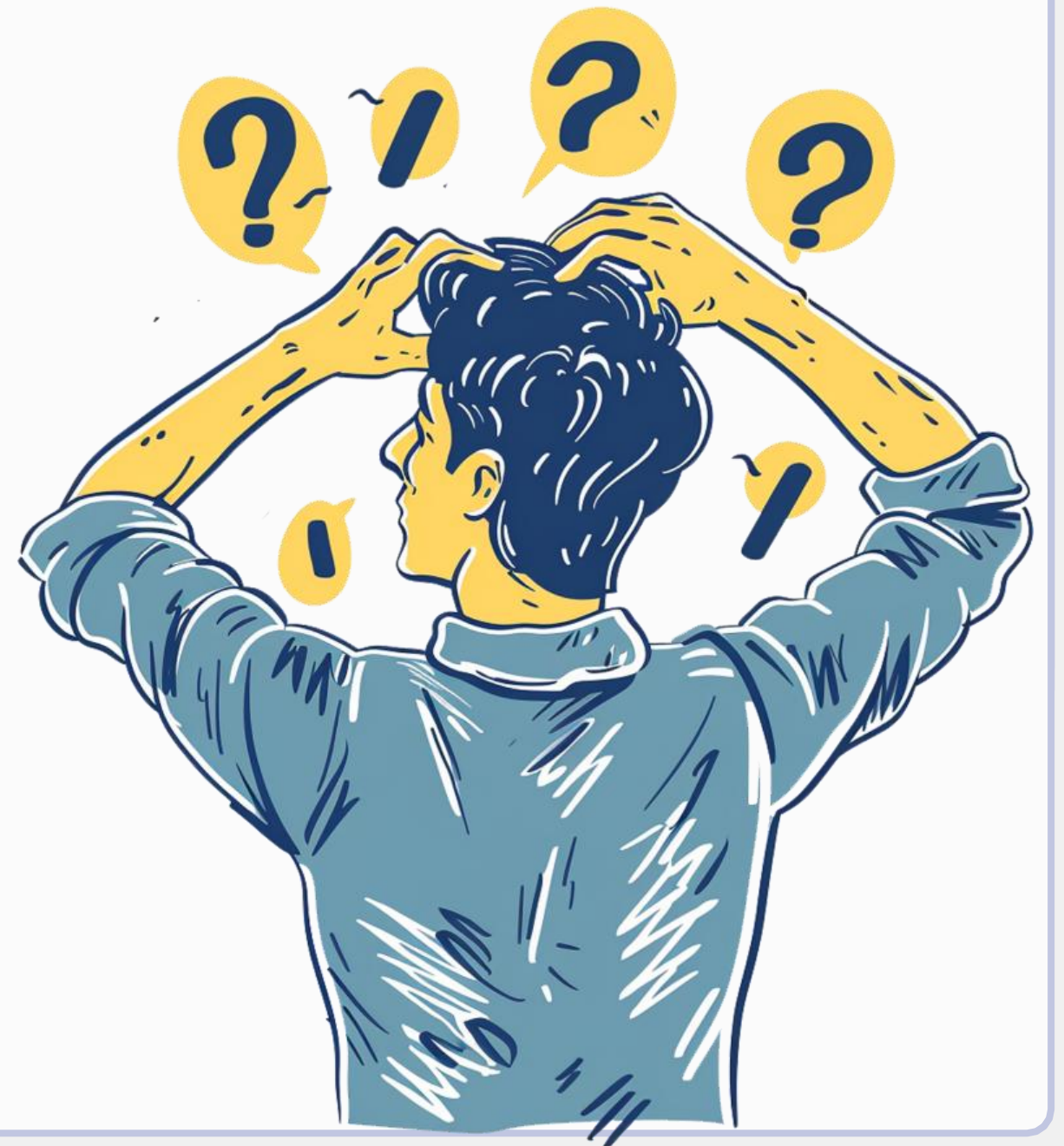
피쳐 제거

하이퍼 파라미터 튜닝

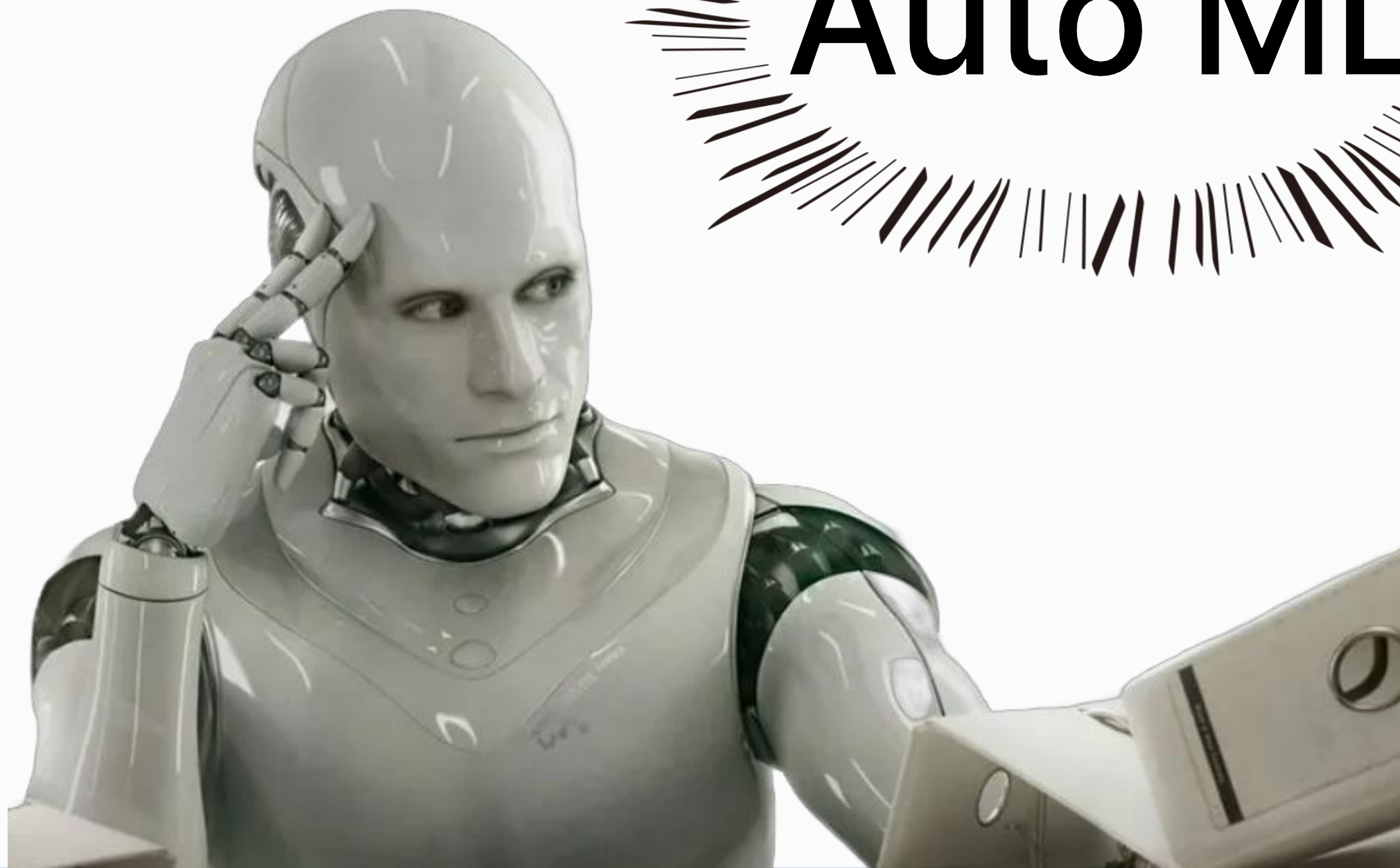
피쳐 생성

학습 모델 변경

데이터 불균형 해소



Auto ML



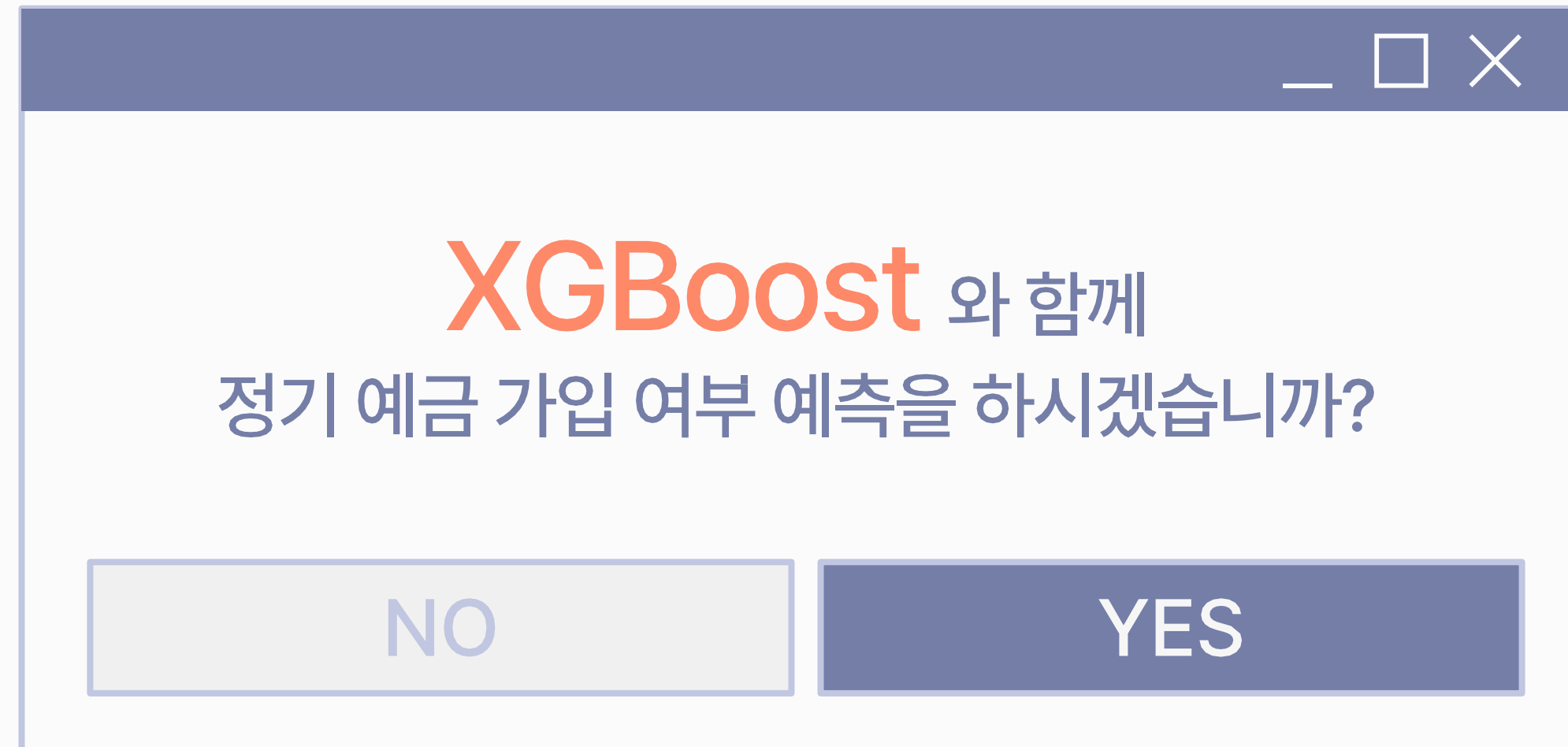
	Model	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC
catboost	CatBoost Classifier	0.9056	0.9306	0.4620	0.6325	0.5937	0.4826	0.4901
lightgbm	Light Gradient Boosting Machine	0.9041	0.9287	0.4639	0.6205	0.5308	0.4786	0.4850
xgboost	Extreme Gradient Boosting	0.9024	0.9226	0.4712	0.6062	0.5302	0.4767	0.4814
lda	Linear Discriminant Analysis	0.8982	0.9100	0.4388	0.5869	0.5021	0.4467	0.4526
gbc	Gradient Boosting Classifier	0.9032	0.9215	0.4064	0.6347	0.4952	0.4446	0.4582
ada	Ada Boost Classifier	0.9004	0.9089	0.3798	0.6214	0.4709	0.4195	0.4353
rf	Random Forest Classifier	0.9023	0.9231	0.3701	0.6433	0.4697	0.4202	0.4398
dt	Decision Tree Classifier	0.8716	0.6947	0.4639	0.4522	0.4576	0.3848	0.3851
nb	Naive Bayes	0.8388	0.8076	0.5461	0.3714	0.4421	0.3519	0.3609
lr	Logistic Regression	0.8991	0.9058	0.3161	0.6393	0.4225	0.3741	0.4023
et	Extra Trees Classifier	0.8958	0.9046	0.3111	0.6060	0.4111	0.3604	0.3846
svm	SVM - Linear Kernel	0.8979	0.8855	0.3003	0.6349	0.4076	0.3595	0.3899
ridge	Ridge Classifier	0.8992	0.9102	0.2786	0.6603	0.3923	0.3473	0.3871
knn	K Neighbors Classifier	0.8864	0.7269	0.2192	0.5352	0.3110	0.2608	0.2919
qda	Quadratic Discriminant Analysis	0.3773	0.5680	0.7383	0.1270	0.2157	0.0214	0.0508
dummy	Dummy Classifier	0.8830	0.5000	0.0000	0.0000	0.0000	0.0000	0.0000

상위 모델

CatBoost

LightGBM

XGBoost



XGBoost 와 함께

정기 예금 가입 여부 예측을 하시겠습니까?

NO

YES



Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

Lv.6

Lv.7

Lv.8

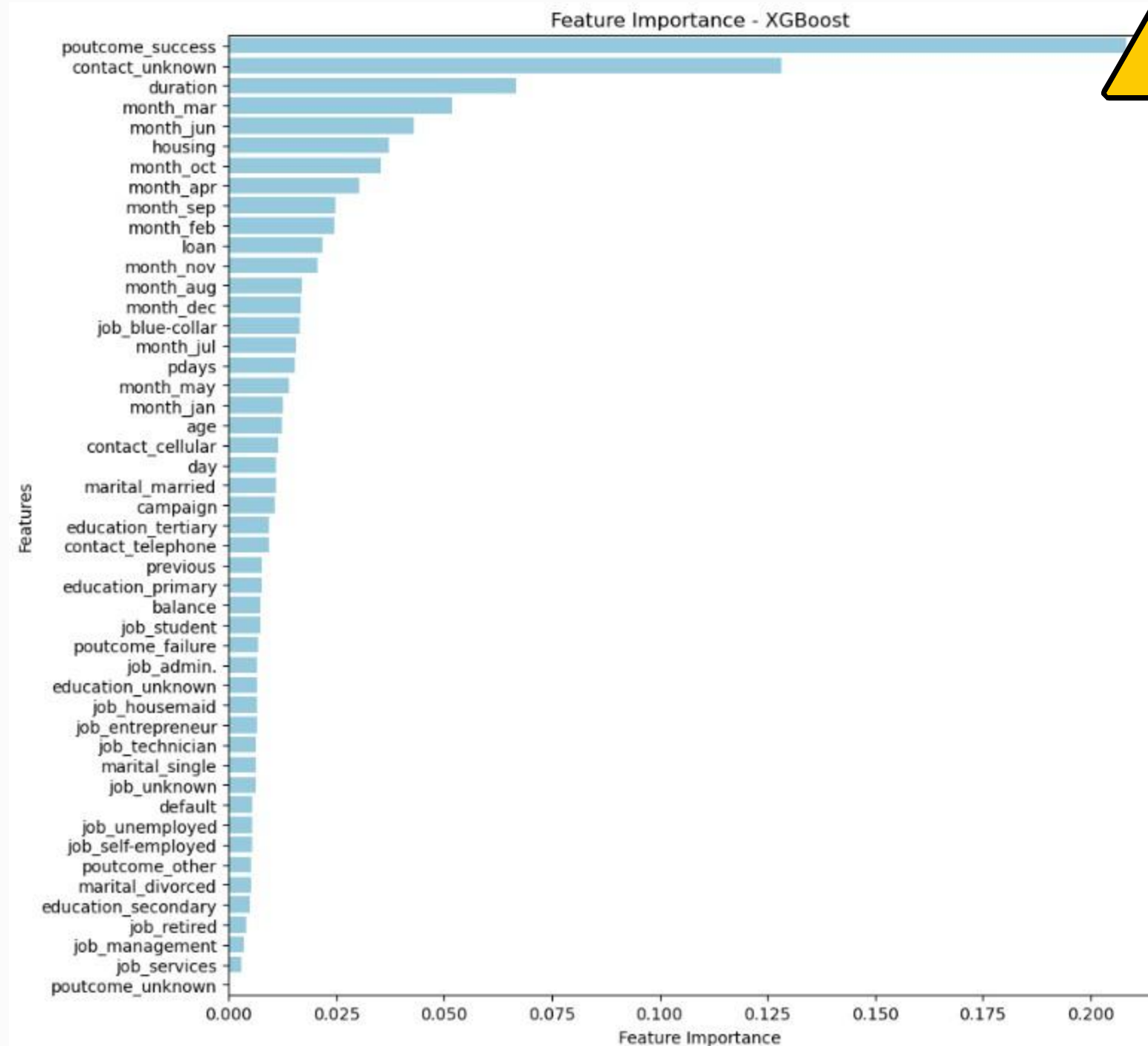
XGBoost

Feature Importance

하나의 **피처 중요도**가 너무 높아
train data 학습 **의존도**가 높음



unseen data 예측에
불리하다고 판단





Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

Lv.6

Lv.7

Lv.8

XGBoost

Feature Engineering

- campaign_days_interaction : 캠페인 연락 횟수 ÷ 최근 연락 후 지난 일 수
- month_contact : 분기 말, 캠페인 연락방법 = 'cellular'
- housing_no : 29살 이하, 60세 이상 주택 대출이 없는 사람
- precam : 이전 캠페인 연락 횟수 < 이번 캠페인 연락 횟수
- good_job : 직업이 'retired', 'student', 'unemployed', 'management'



Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

Lv.6

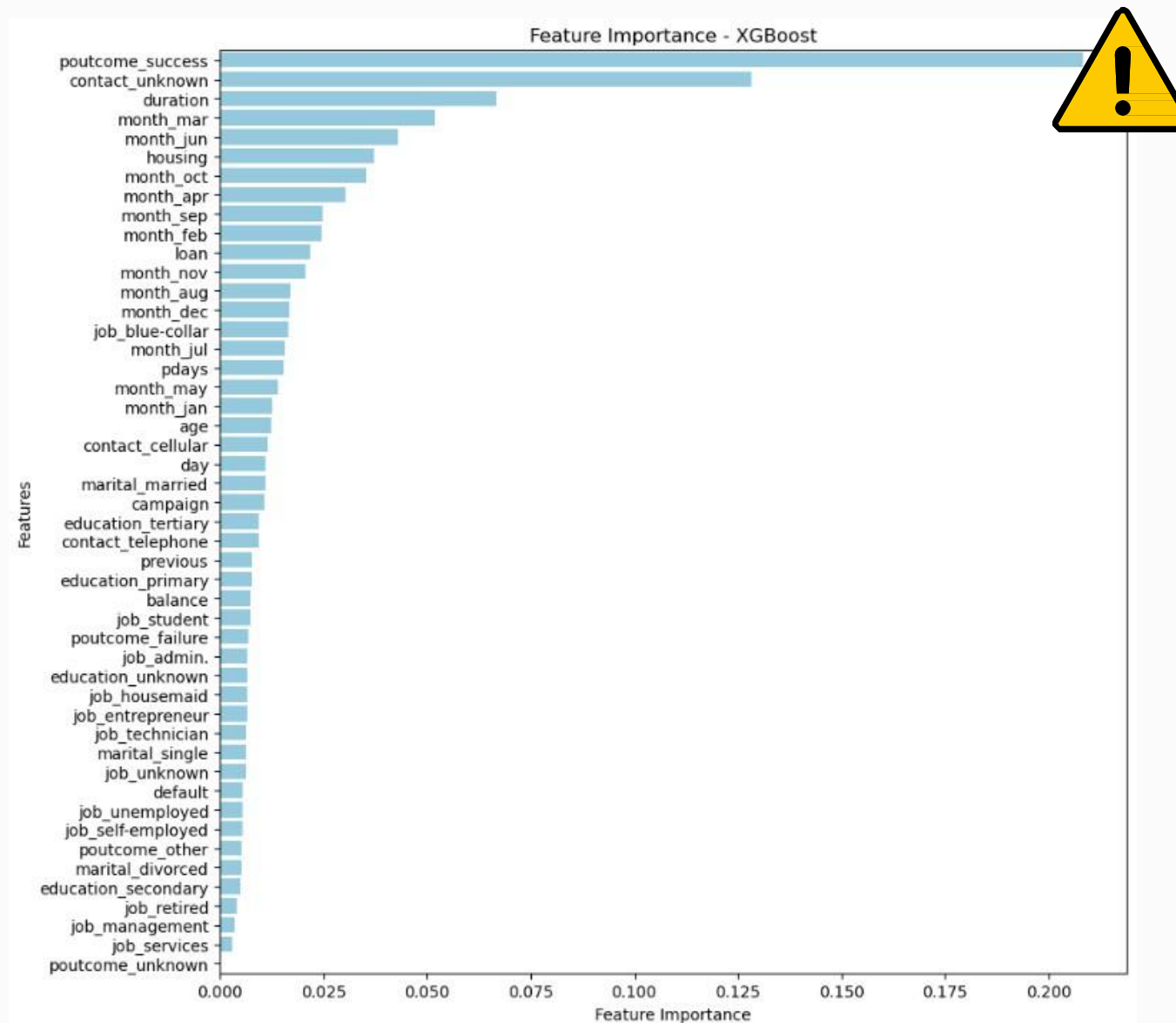
Lv.7

Lv.8

XGBoost

Feature Selection

피쳐 중요도가 **고르게 분포**하는 방향으로 피쳐 제거



피쳐 중요도가 **0에 수렴**하는 변수는
예측 성능을 저해할 수 있다고 판단



Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

Lv.6

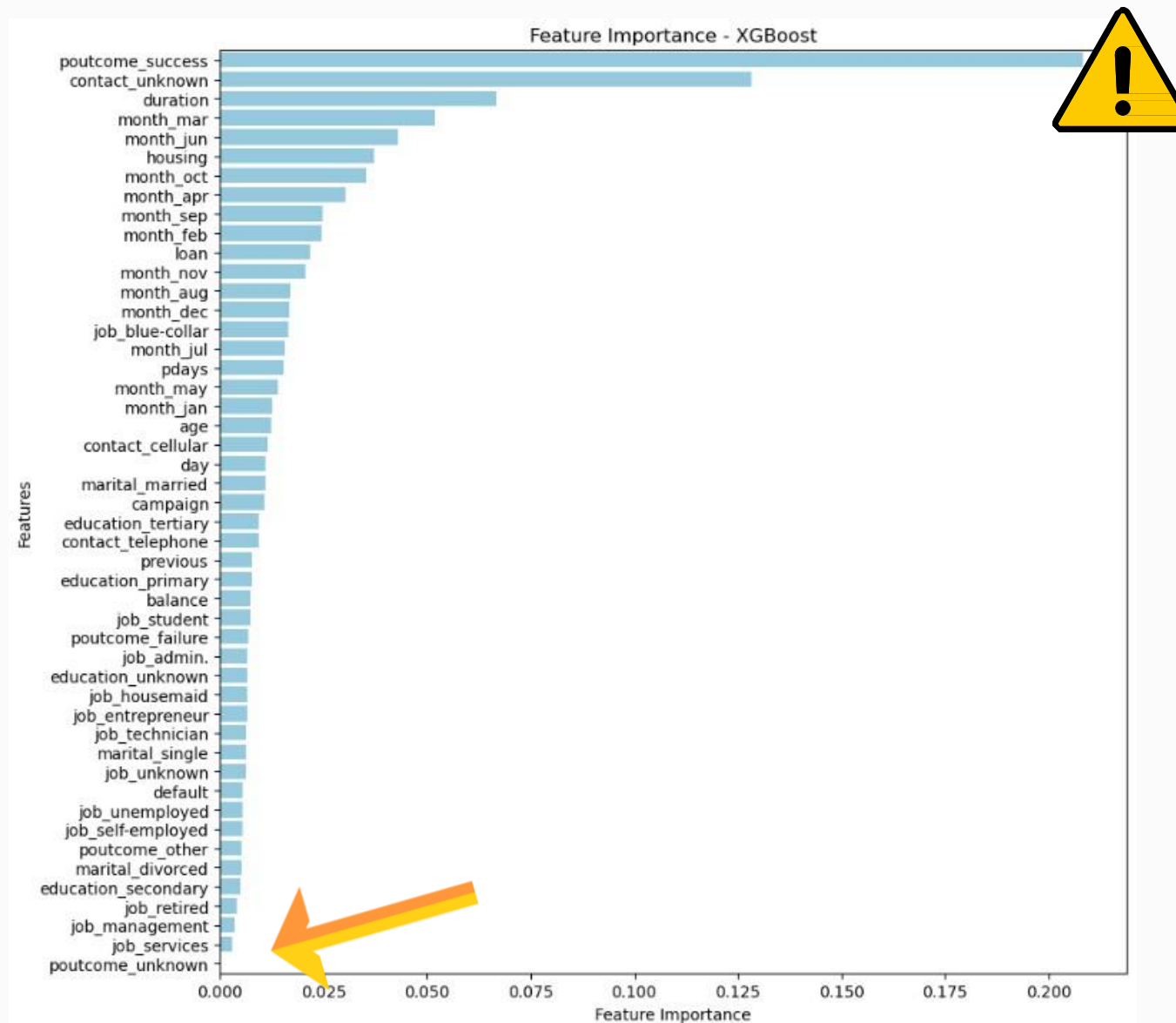
Lv.7

Lv.8

XGBoost

Feature Selection

피처 중요도가 **고르게 분포**하는 방향으로 피처 제거



피처 중요도가 **0에 수렴**하는 변수는
예측 성능을 저해할 수 있다고 판단



Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

Lv.6

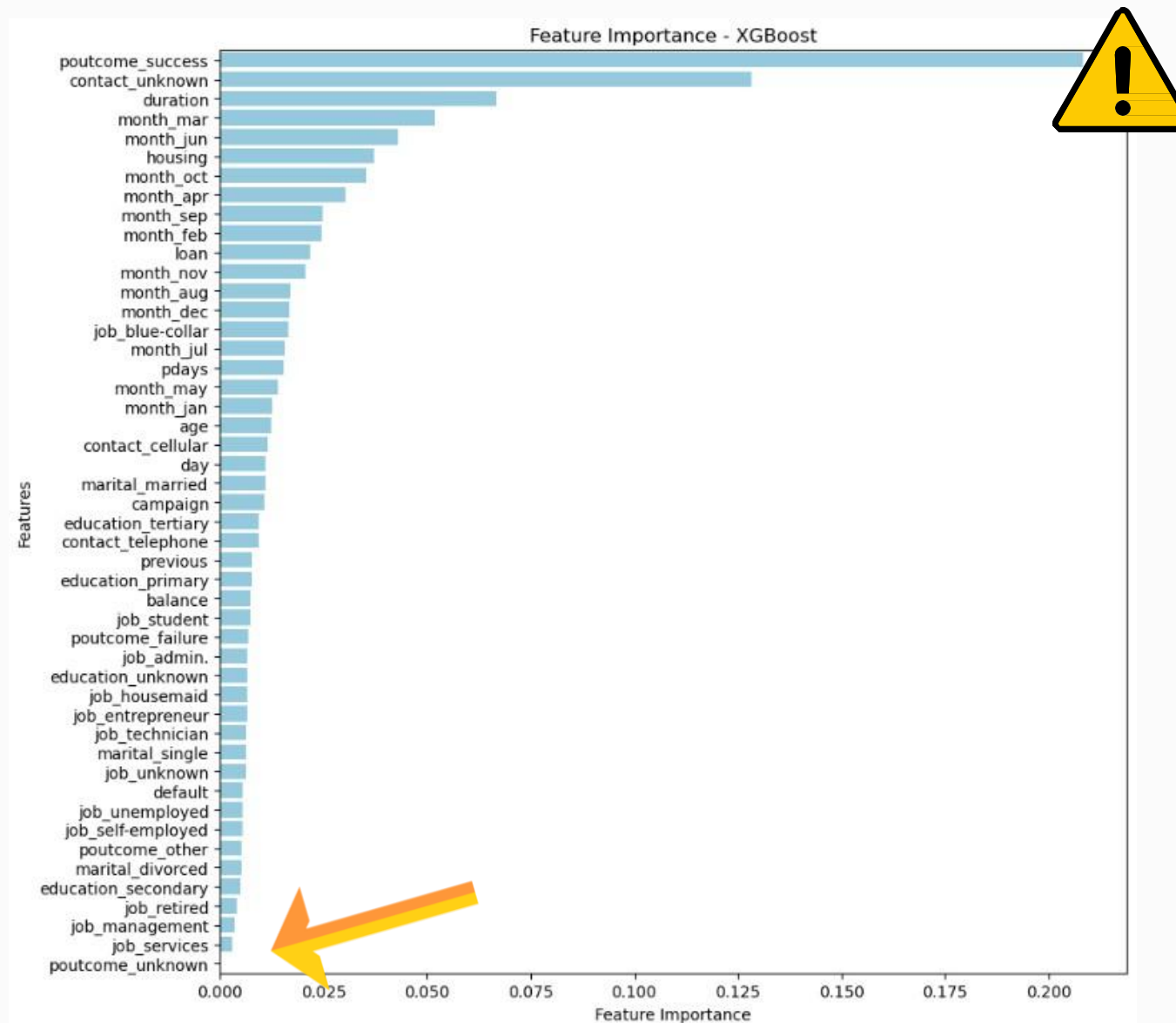
Lv.7

Lv.8

XGBoost

Feature Selection

피처 중요도가 **고르게 분포**하는 방향으로 피처 제거



피처 중요도가 **0에 수렴**하는 변수는
예측 성능을 저해할 수 있다고 판단



poutcome_unknown 제거



Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

Lv.6

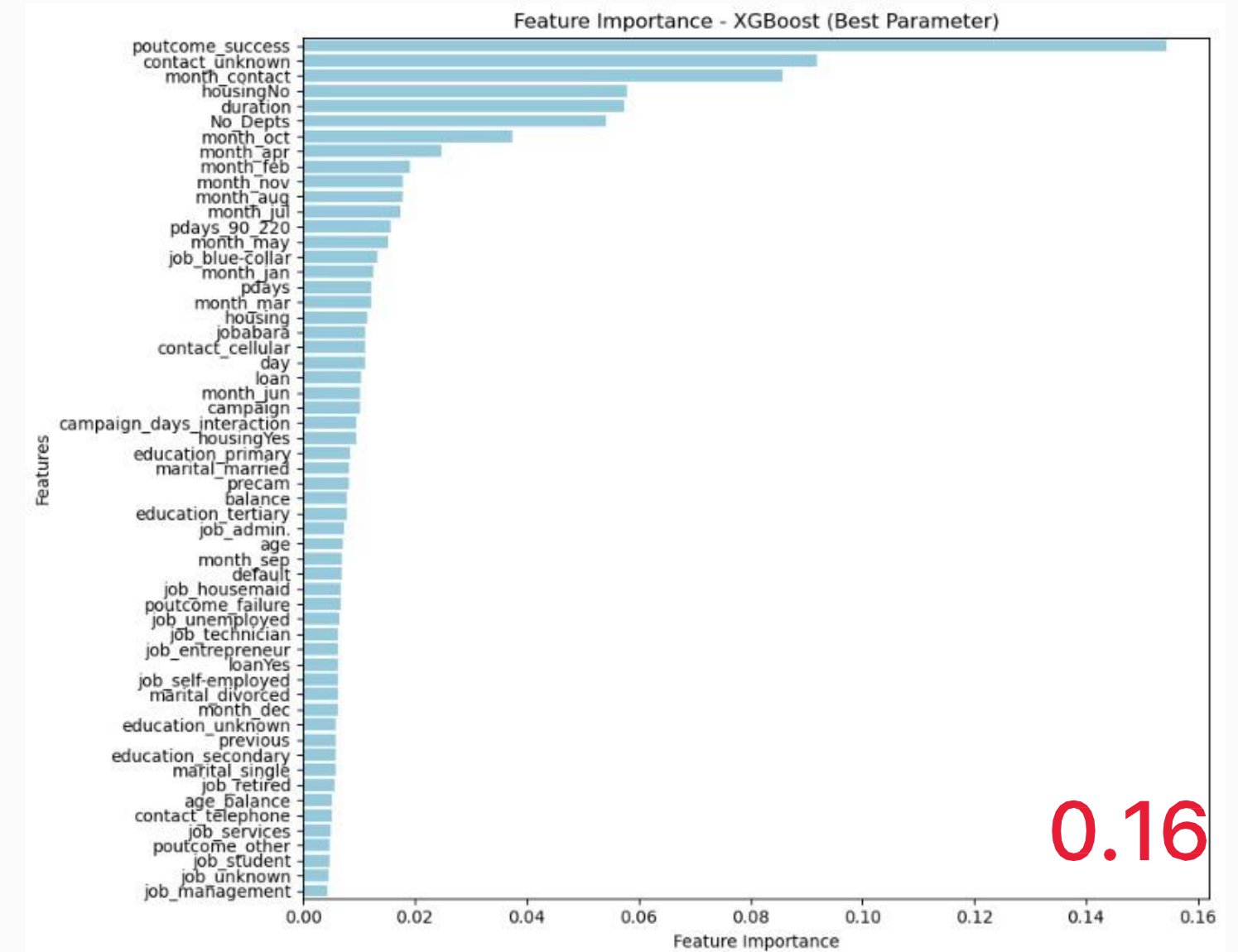
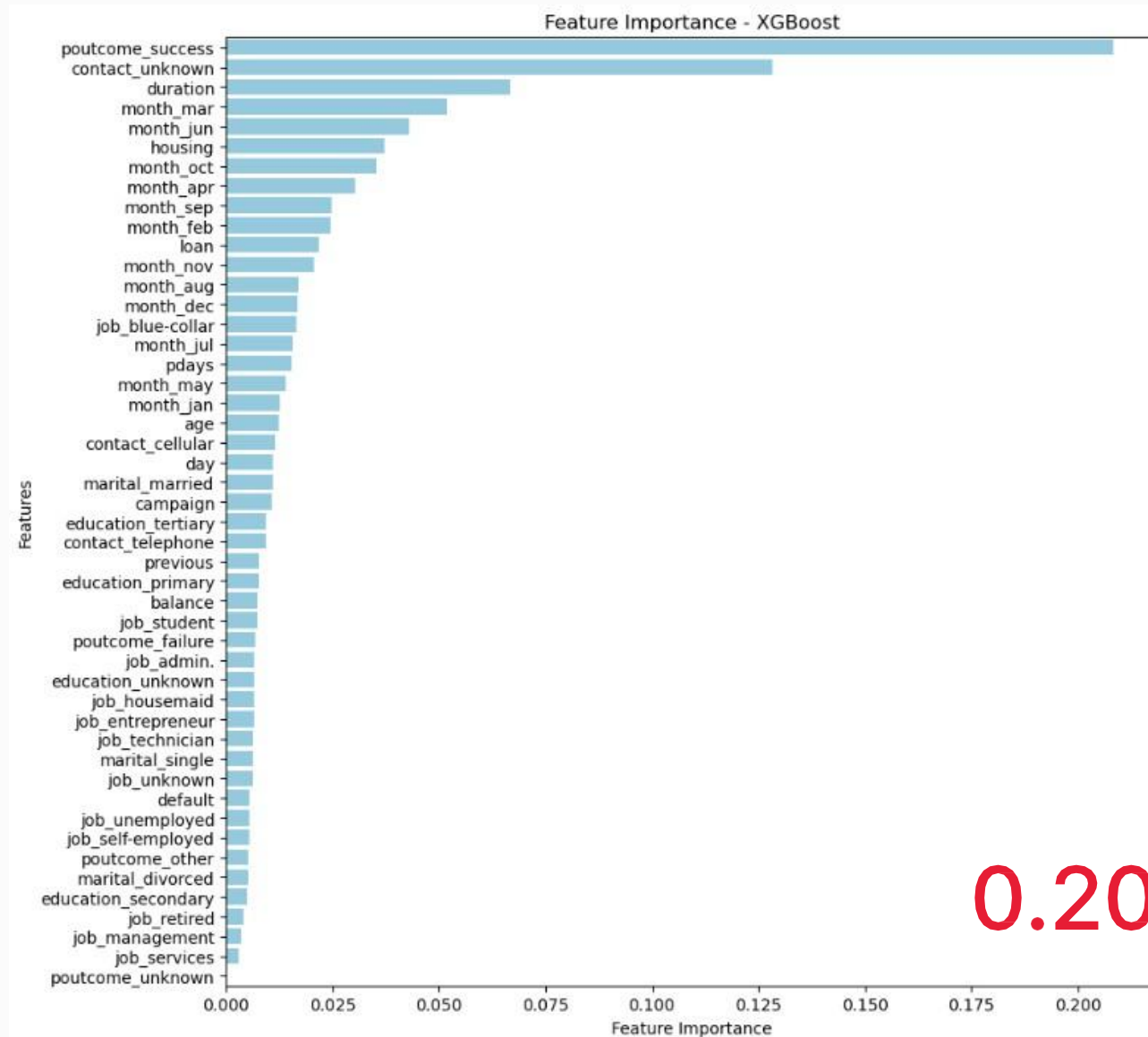
Lv.7

Lv.8

XGBoost

Feature Importance

피쳐 중요도가 **고르게 분포**





Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

Lv.6

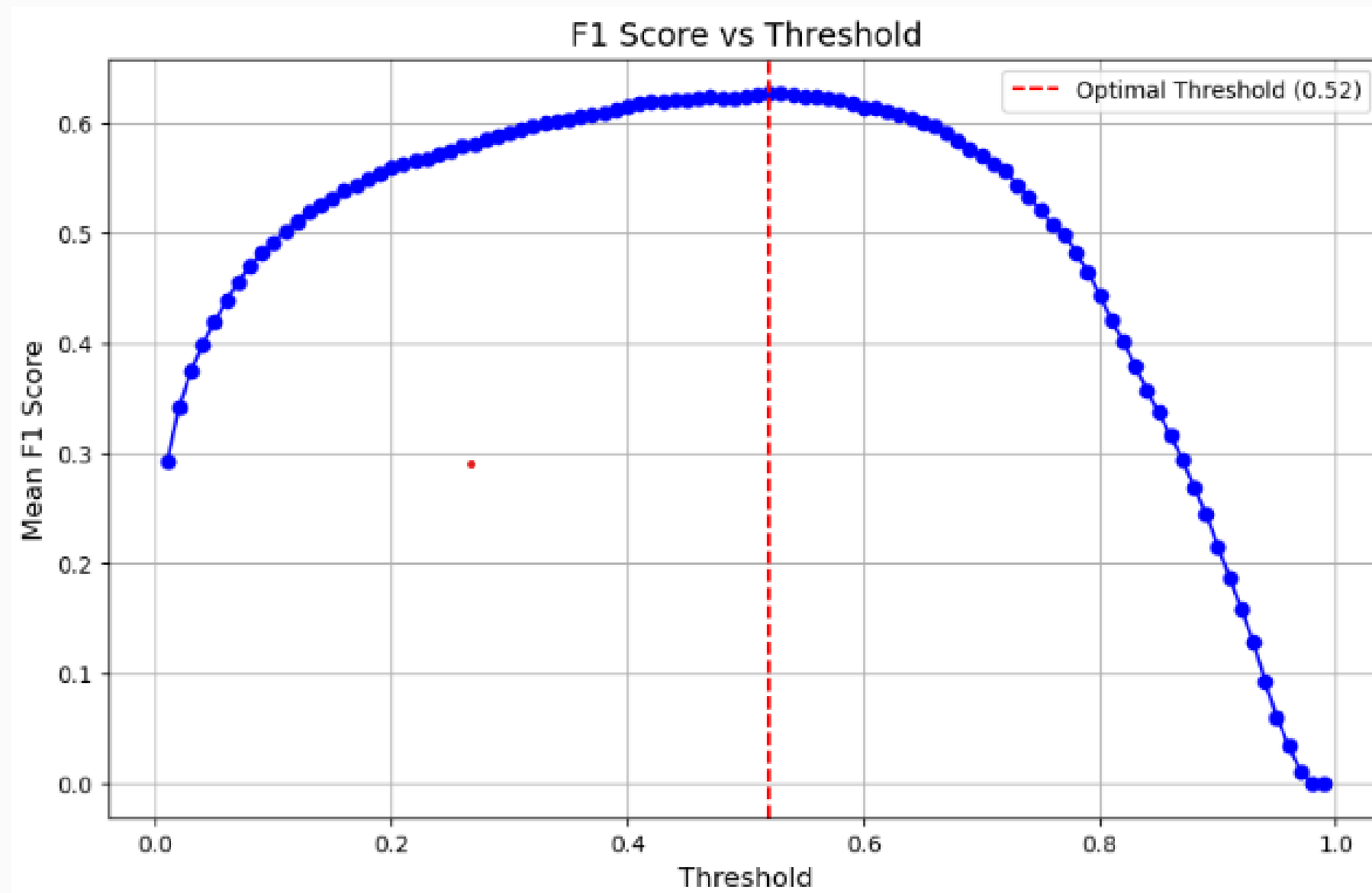
Lv.7

Lv.8

XGBoost

Optimize Threshold

Cross-Validation을 활용해 **최적의 임계값** 설정



임계값이 **0.52**일 때
F1 Score가 가장 높음

LightGBM

와 함께

정기 예금 가입 여부 예측을 하시겠습니까?

NO

YES



Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

Lv.6

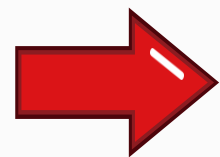
Lv.7

Lv.8

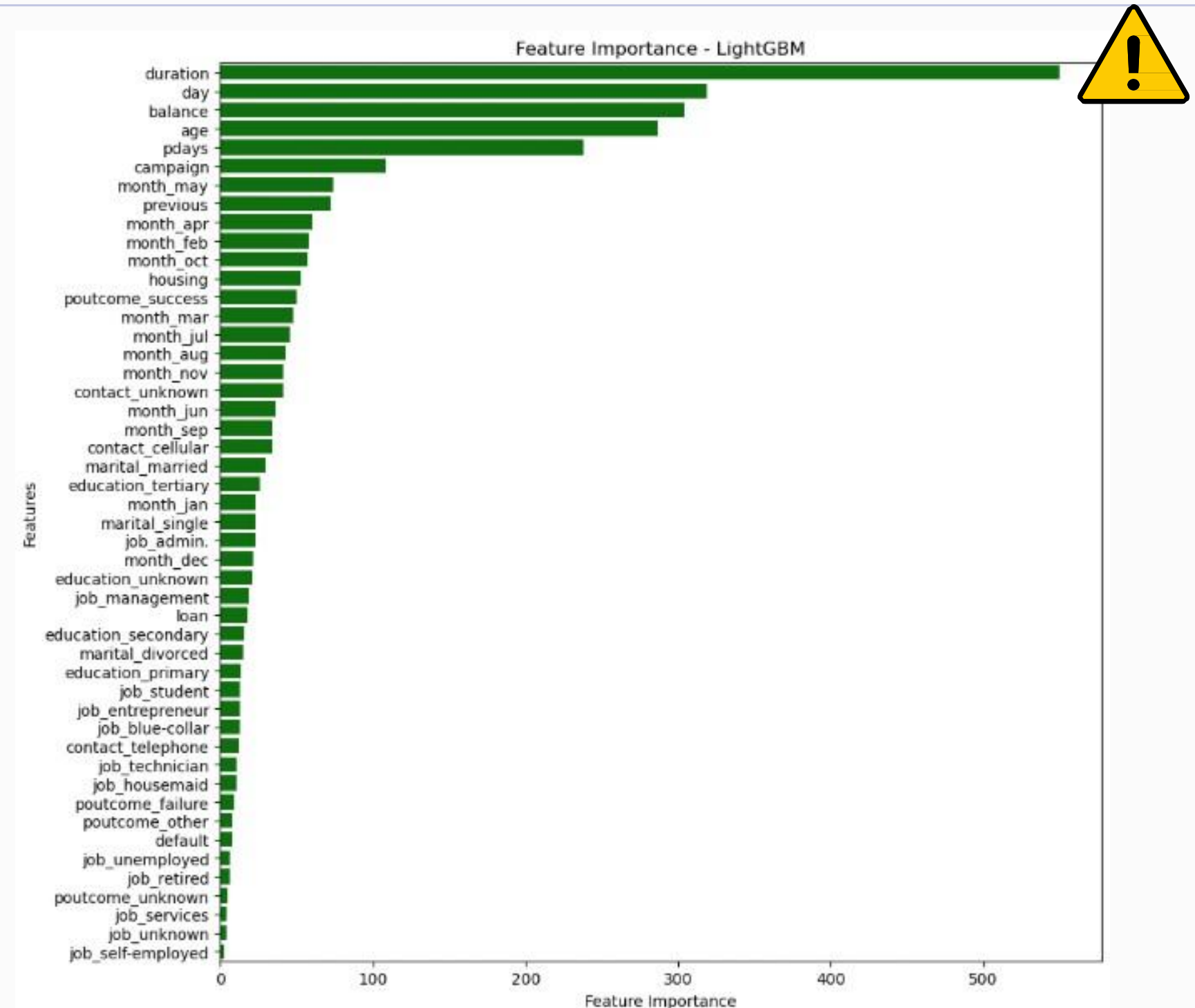
LightGBM

Feature Importance

하나의 **피쳐 중요도**가 너무 높아
train data 학습 **의존도**가 높음



unseen data 예측에
불리하다고 판단





Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

Lv.6

Lv.7

Lv.8

LightGBM

Feature Engineering

- No_Debts : 채무 불이행 여부, 개인/주택 대출 모두 **없음**
- month_contact : 분기 **말**, 캠페인 연락방법 = '**cellular**'
- con_du : 캠페인 연락방법이 '**unknown**'이며 통화시간이 **517초 이하**

XGBoost 모델에서는 좋은 성능을 보였던 피처들이
LightGBM에서는 **미미한 효과**를 보여 위 **3개 피처**만 추가



LightGBM

Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

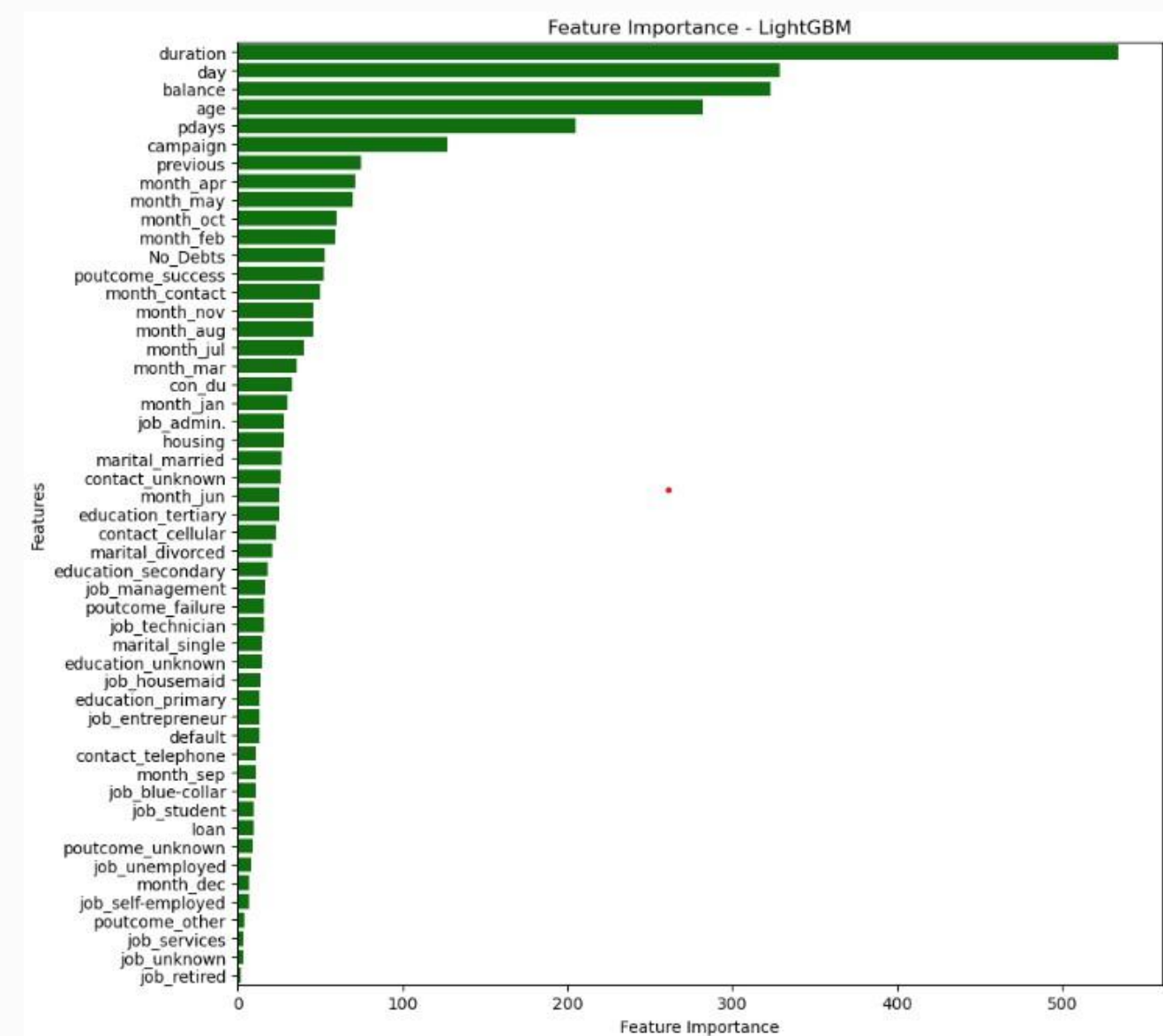
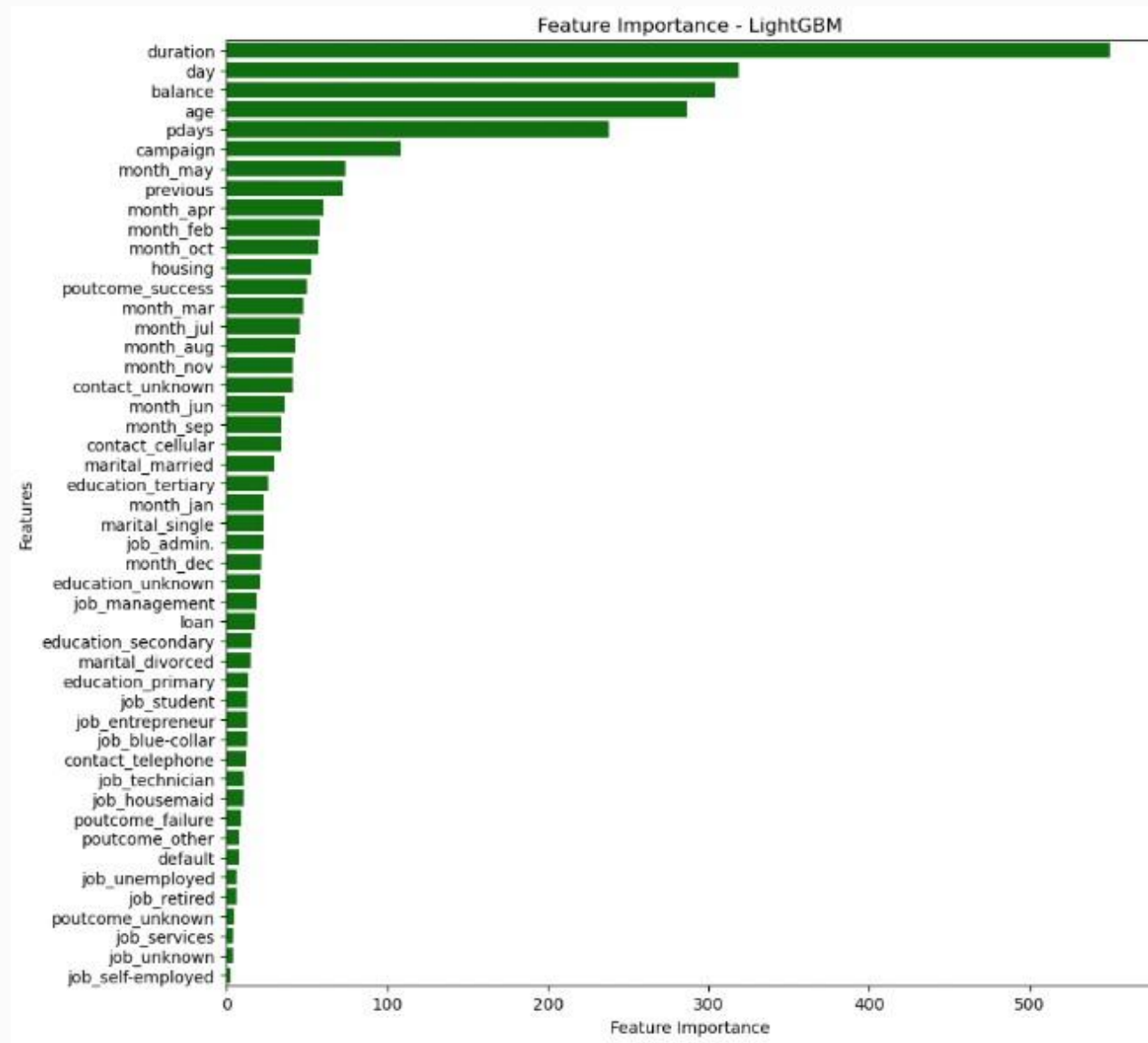
Lv.6

Lv.7

Lv.8

Feature Importance

피처 중요도가 **고르게 분포**





LightGBM

Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

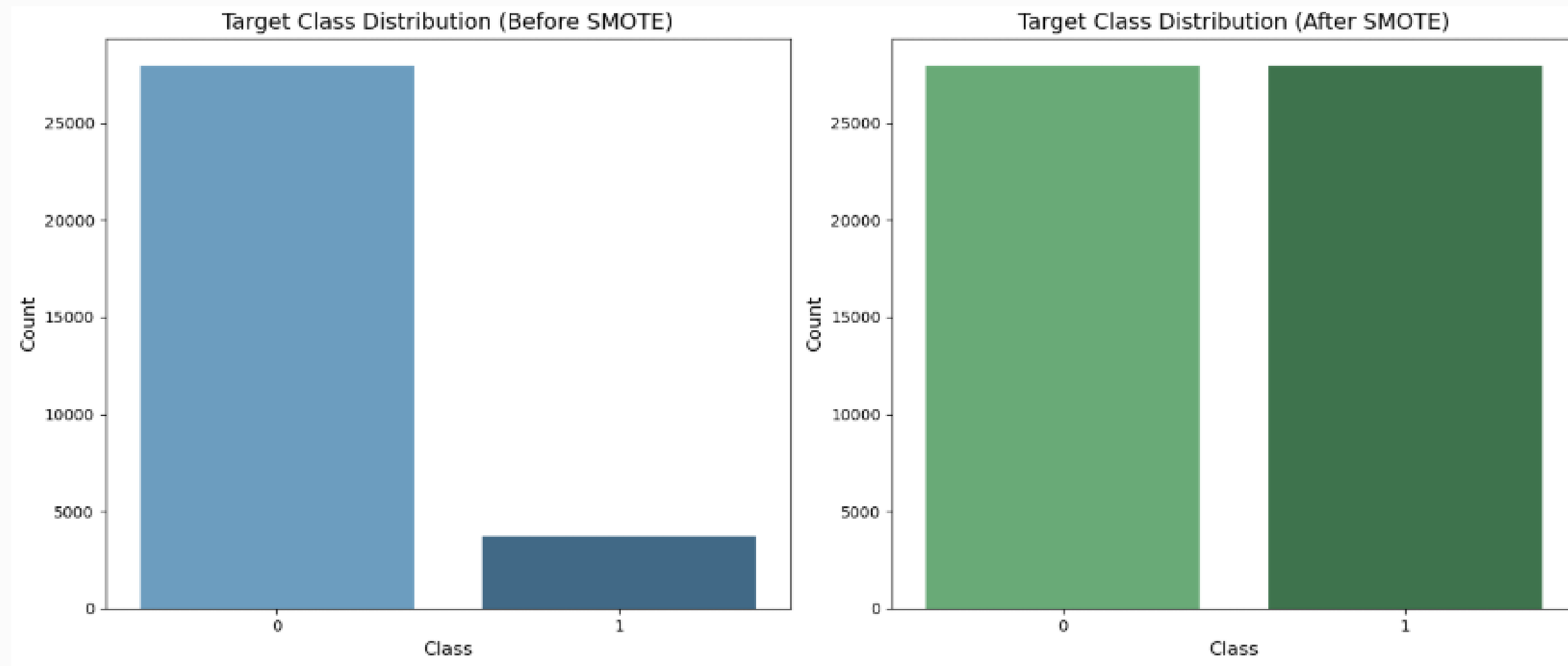
Lv.6

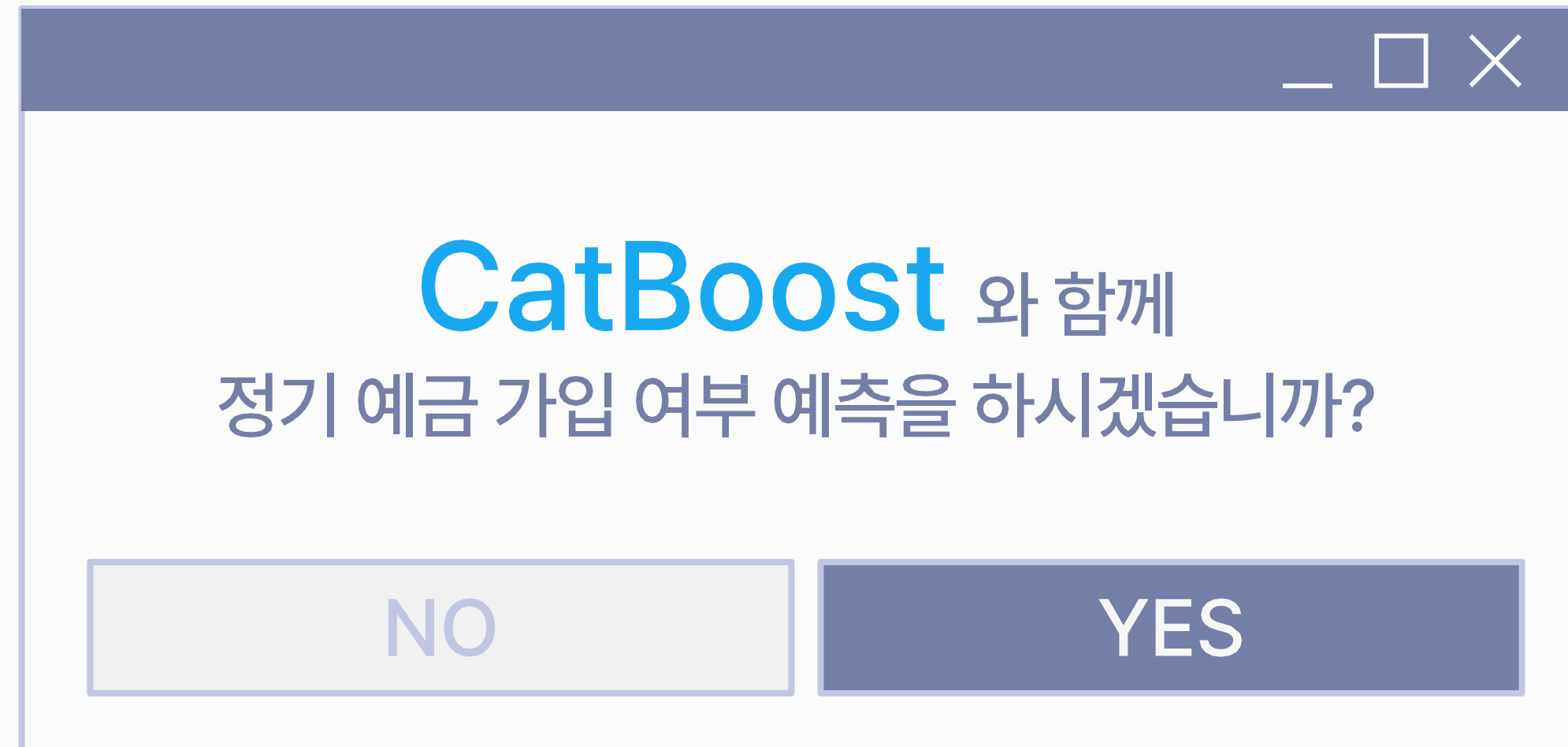
Lv.7

Lv.8

OverSampling

임계값 조절 대신 **SMOTE**를 활용하여 **불균형 해소**





CatBoost 와 함께

정기 예금 가입 여부 예측을 하시겠습니까?

NO

YES



Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

Lv.6

Lv.7

Lv.8

CatBoost

범주형 변수 효율적 처리 → 다양한 시도 가능



Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

Lv.6

Lv.7

Lv.8

CatBoost

범주형 변수 효율적 처리 → 다양한 시도 가능

변수 + ' / ' + 변수

변수 자체의 성질을 살리며 새로운 피처 생성



Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

Lv.6

Lv.7

Lv.8

CatBoost

범주형 변수 효율적 처리 ➡ 다양한 시도 가능

- ['poutcome_balance'] = ['poutcome'] + '/' + ['balance_cat']
✱ 'balance'를 4분위수를 기준으로 구간화 후, 'poutcome' 변수와 결합
- ['contact_duration'] = ['contact'] + '/' + ['duration_category']
✱ 'duration'을 4분위수를 기준으로 구간화 후, 'contact' 변수와 결합



CatBoost

Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

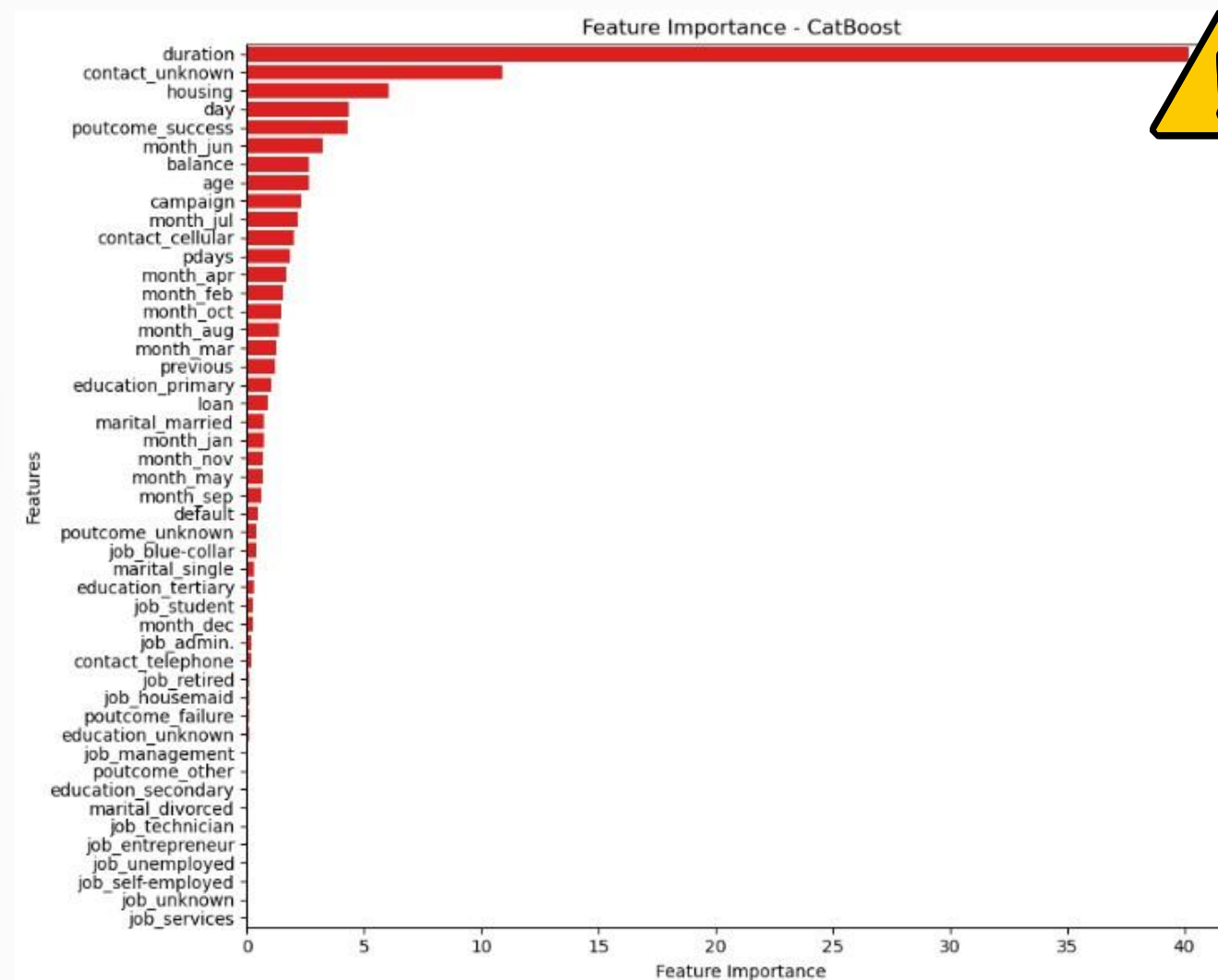
Lv.6

Lv.7

Lv.8

Feature Selection

피쳐 중요도가 **고르게 분포**하는 방향으로 피쳐 제거



various_variants	0.01
education_unknown	0.01
marital_divorced	0.01
job_admin.	0.00
poutcome_failure	0.00
job_self-employed	0.00
job_unknown	0.00
month_aug	0.00
month_jan	0.01



CatBoost

Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

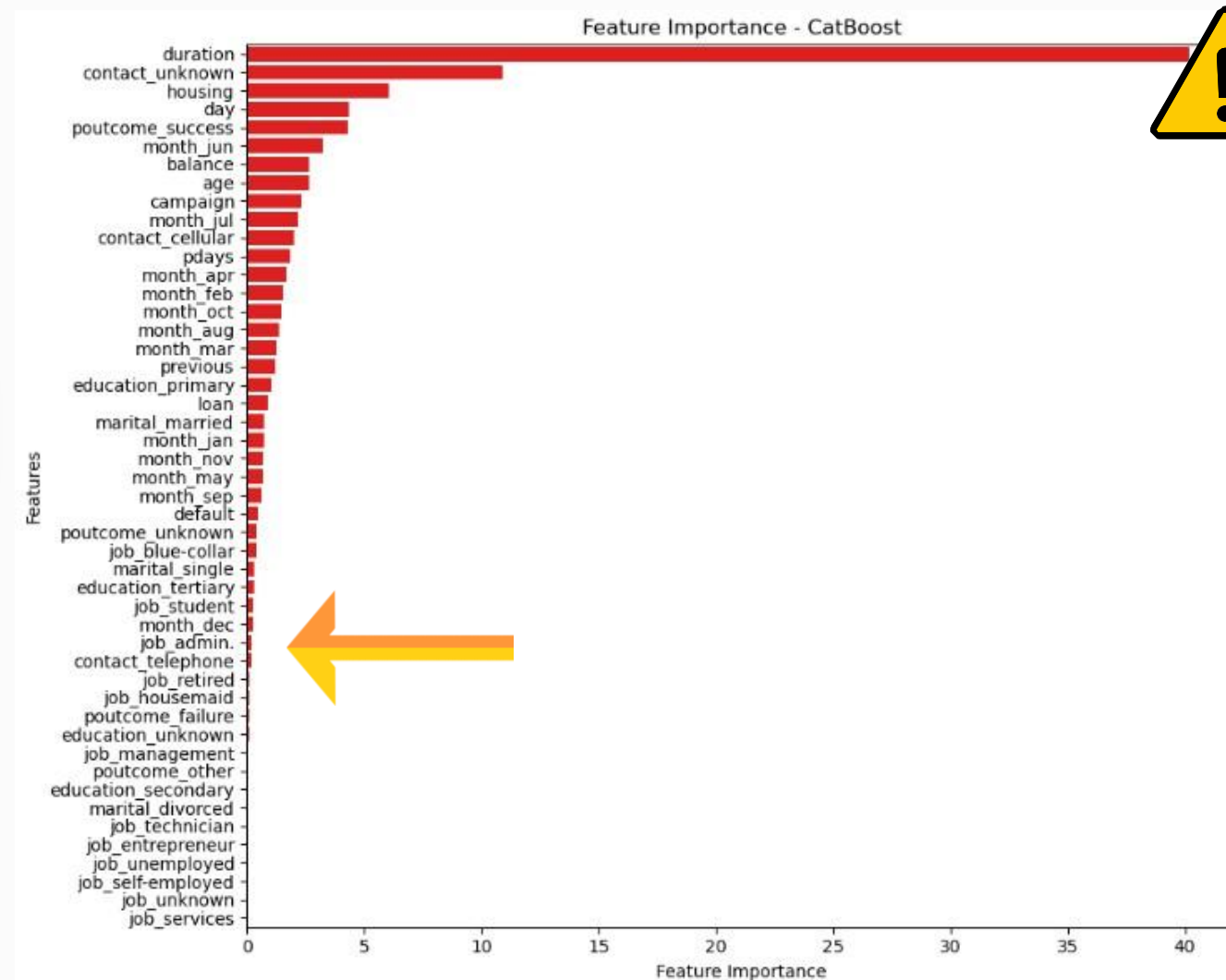
Lv.6

Lv.7

Lv.8

Feature Selection

피쳐 중요도가 **고르게 분포**하는 방향으로 피쳐 제거



education_unknown	0.01
marital_divorced	0.01
job_admin.	0.00
poutcome_failure	0.00
job_self-employed	0.00
job_unknown	0.00
month_aug	0.00
month_jan	0.01



CatBoost

Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

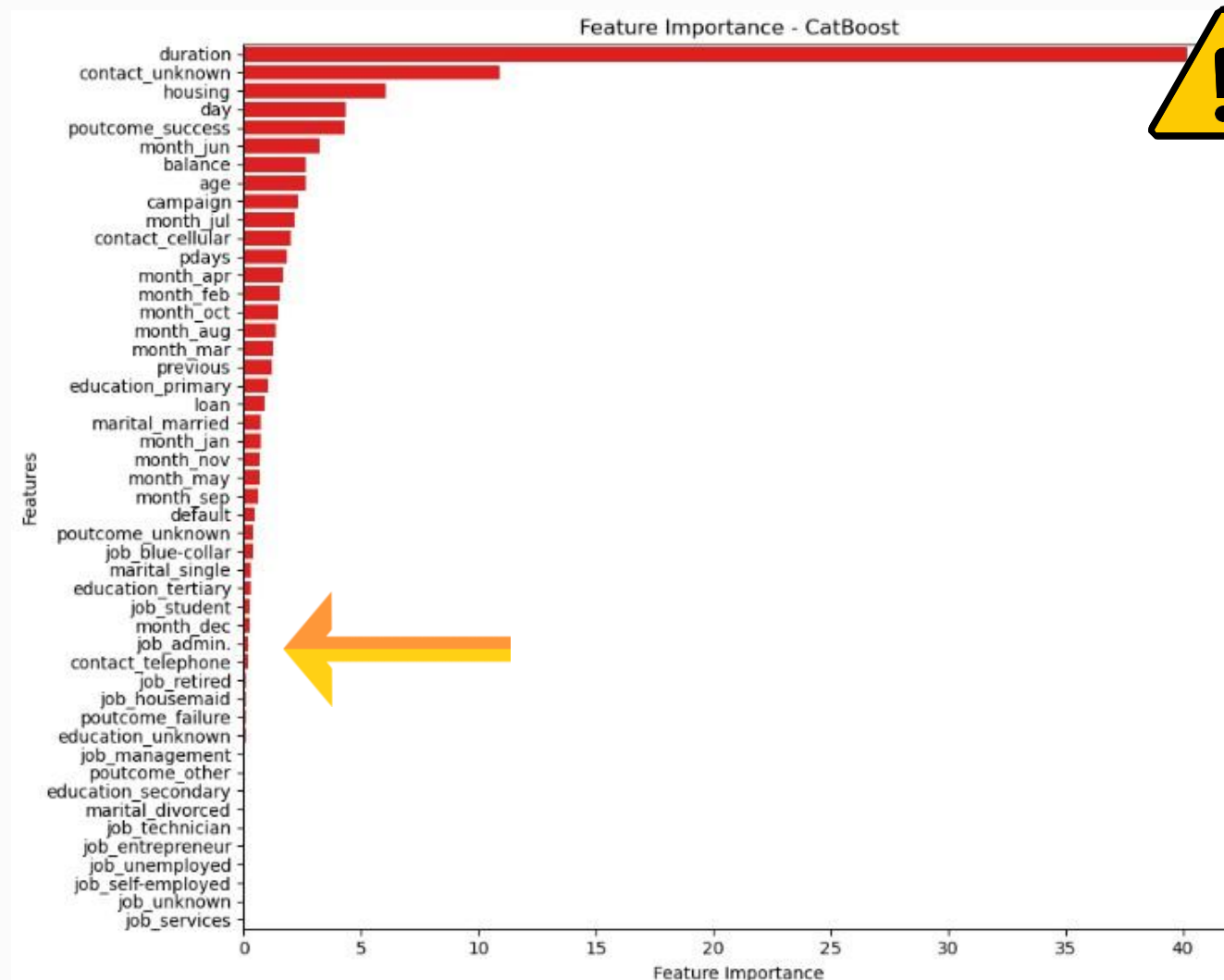
Lv.6

Lv.7

Lv.8

Feature Selection

피쳐 중요도가 **고르게 분포**하는 방향으로 피쳐 제거



education_unknown	0.01
marital_divorced	0.01
job_admin.	0.00
poutcome_failure	0.00
job_self-employed	0.00
job_unknown	0.00
month_aug	0.00
month_sep	0.01

'job_admin.' 변수의
피어슨 상관계수와 피쳐중요도
둘 다 낮으므로 **제거**하기로 판단



Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

Lv.6

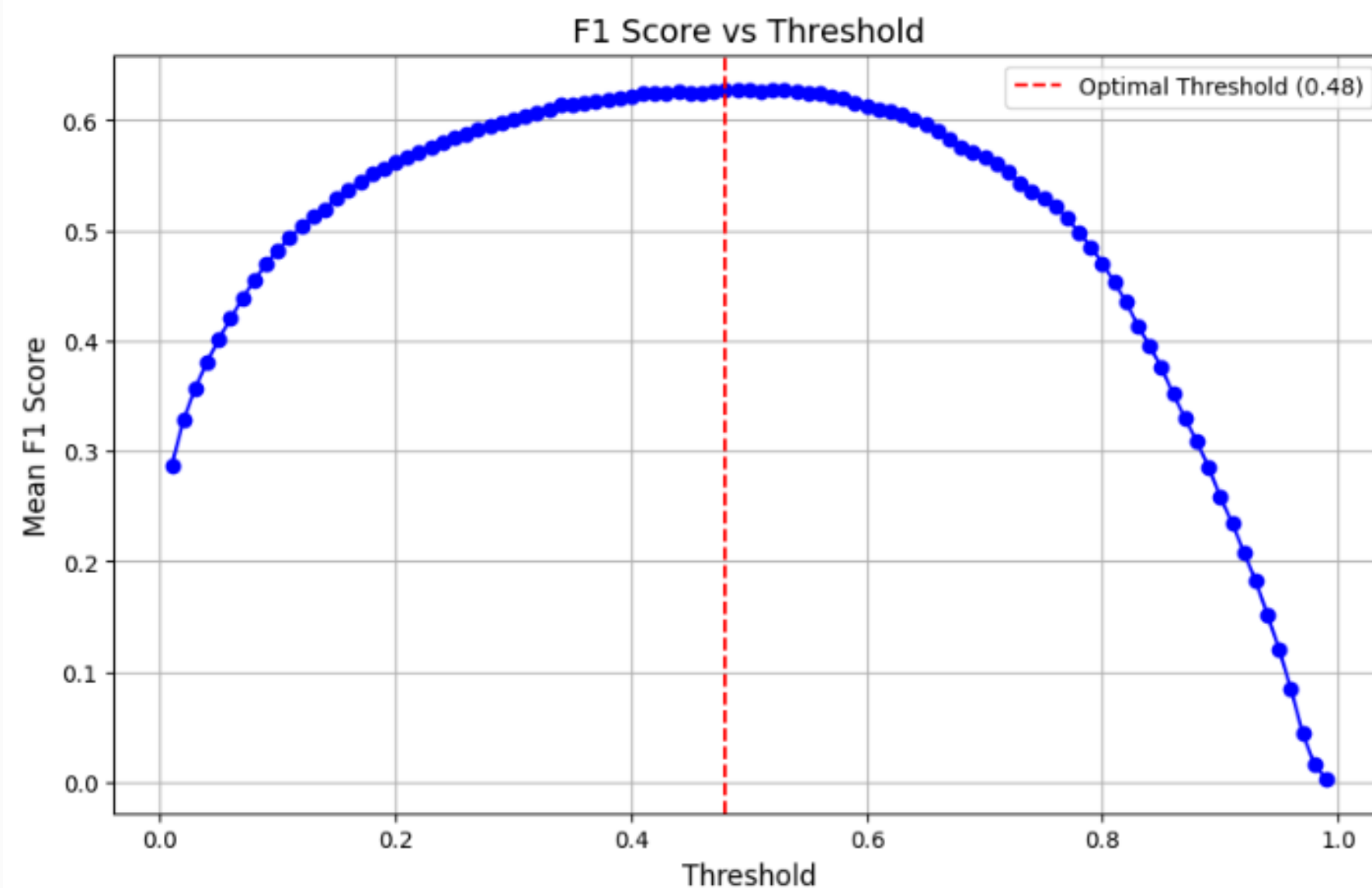
Lv.7

Lv.8

CatBoost

Optimize Threshold

Cross-Validation을 활용해 **최적의 임계값** 설정



임계값이 **0.48**일 때
F1 Score가 가장 높음



CatBoost

Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

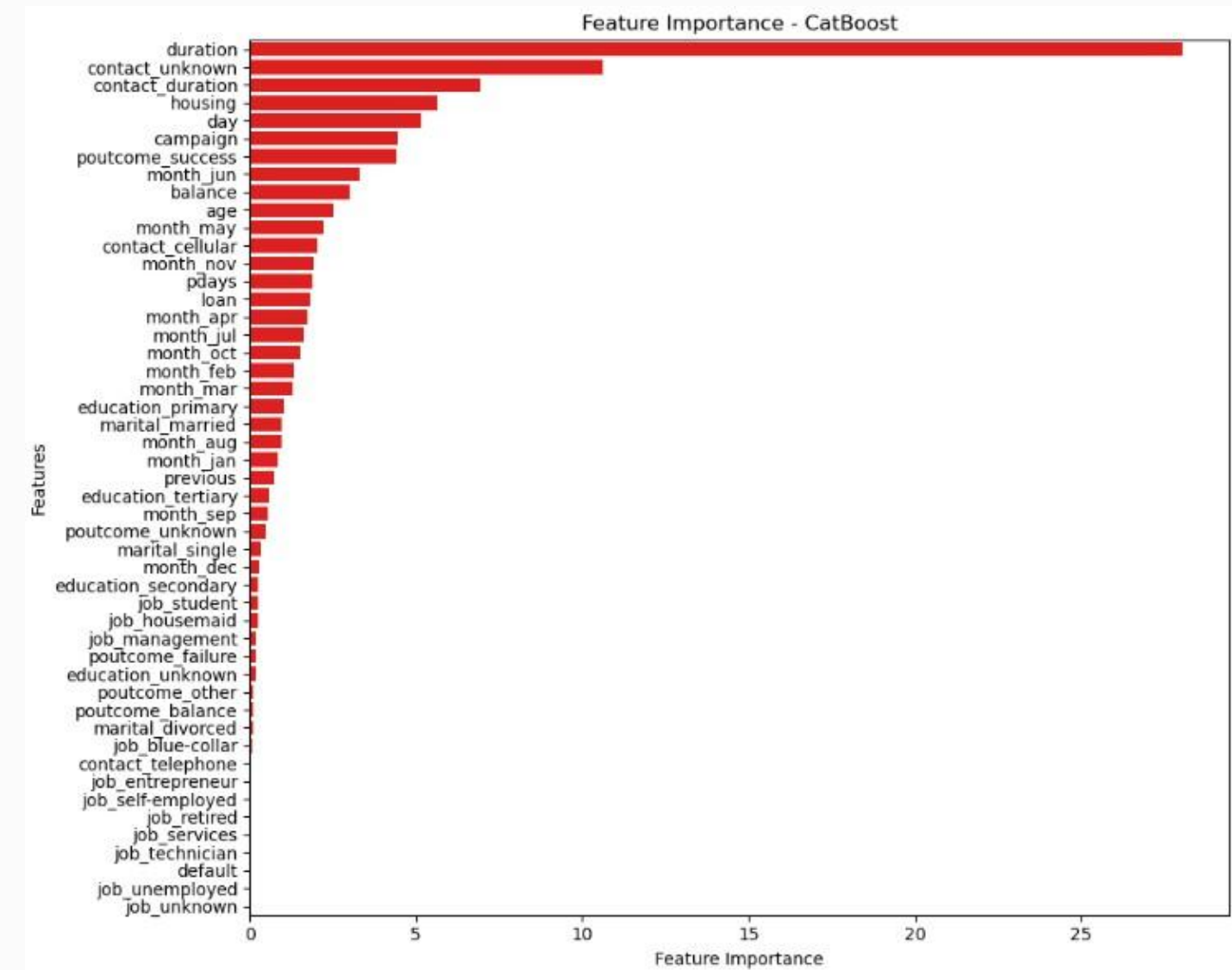
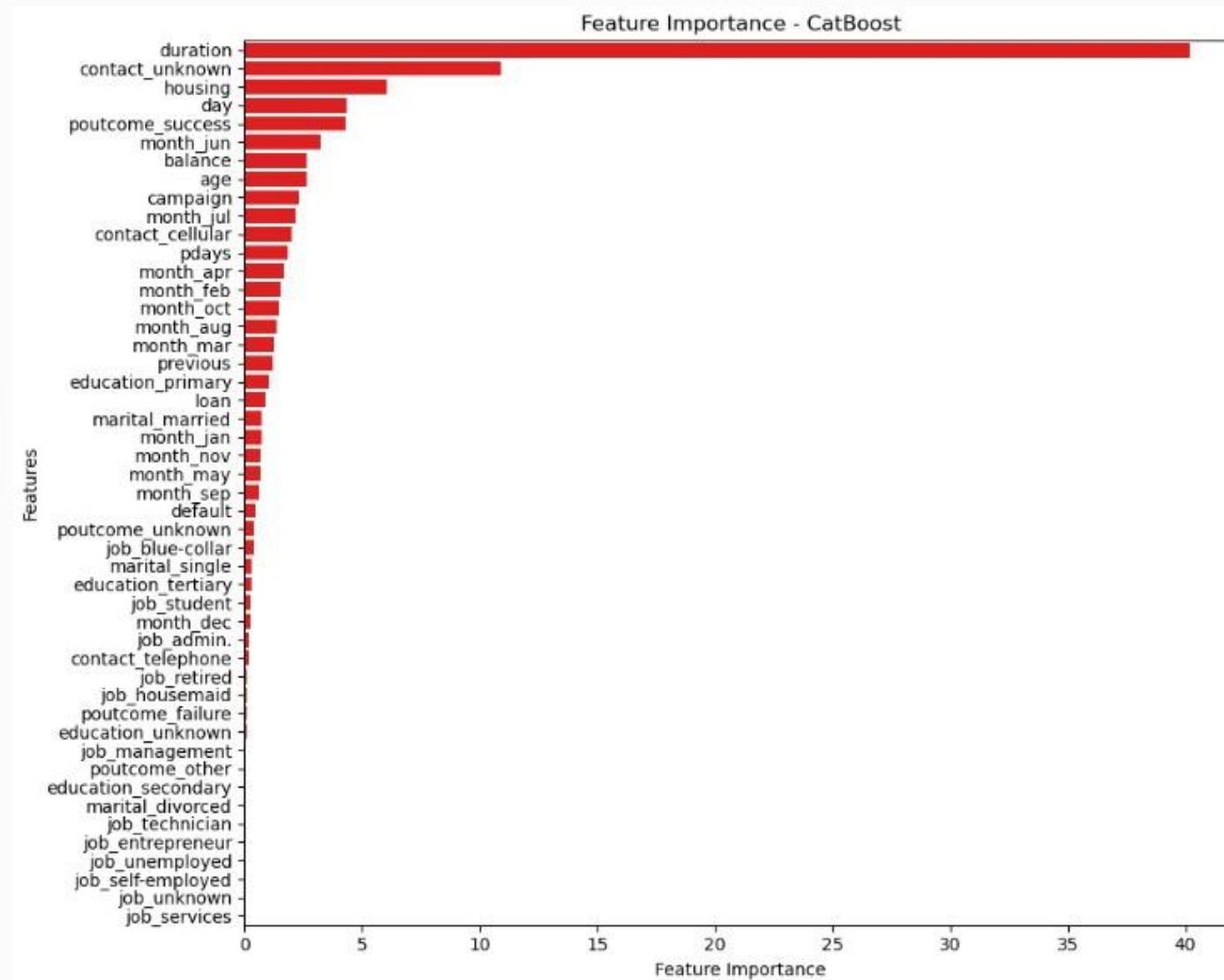
Lv.6

Lv.7

Lv.8

Feature Importance

피쳐 중요도가 **고르게 분포**





Lv.1

Lv.2

Lv.3

Lv.4

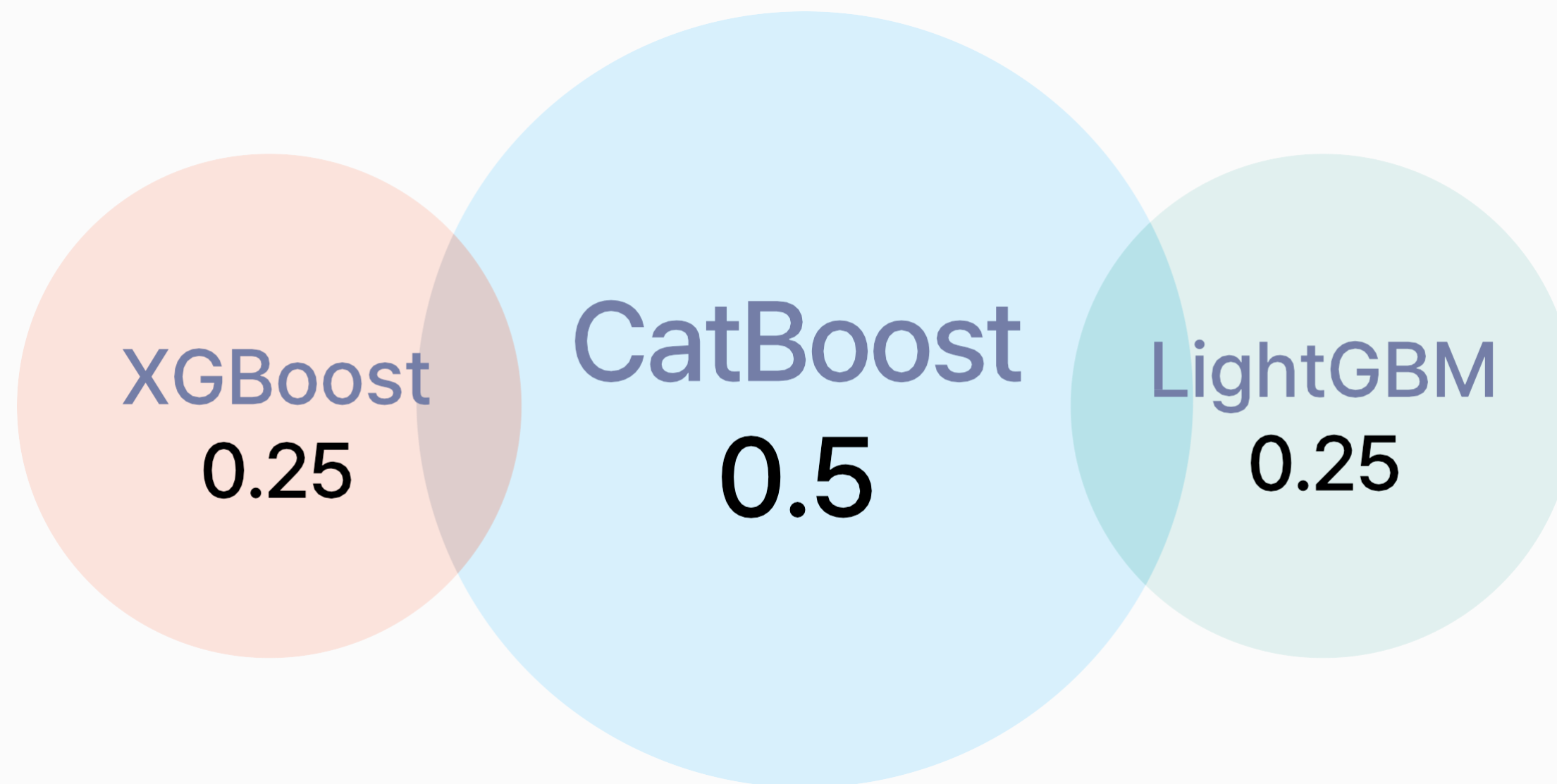
Lv.5

Lv.6

Lv.7

Lv.8

Submission Ensemble



1 : 2 : 1



Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

Lv.6

Lv.7

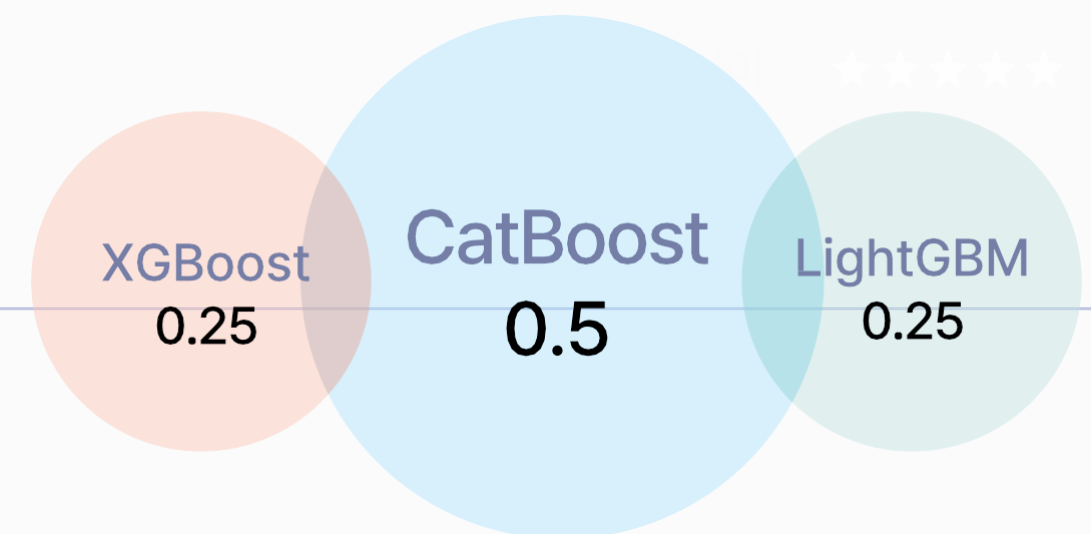
Lv.8

Submission Ensemble

Threshold Control

[CASE 1] : 임계값을 **0.5**로 설정

CatBoost가 **1**로 예측





Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

Lv.6

Lv.7

Lv.8

Submission Ensemble

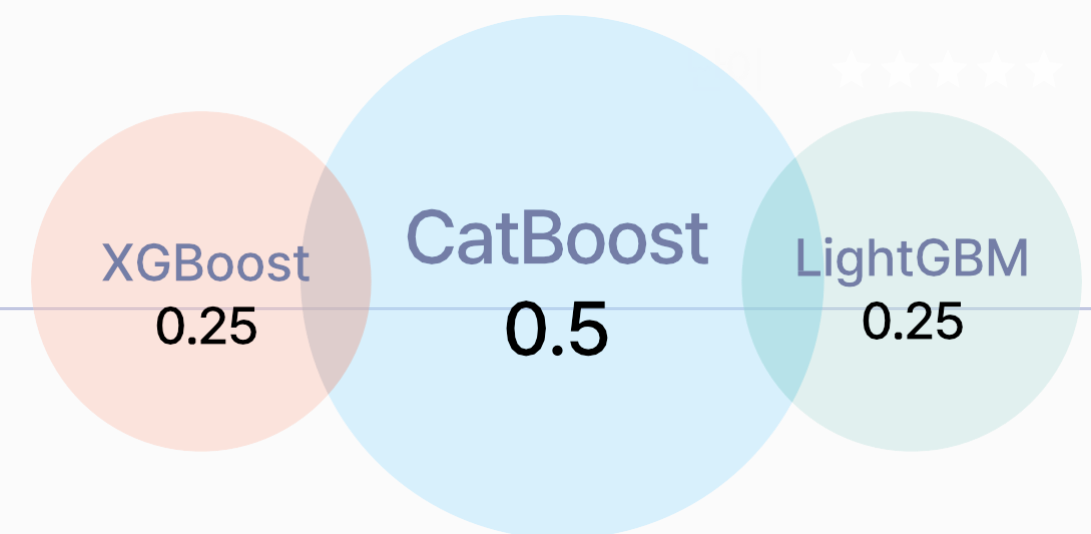
Threshold Control

[CASE 1] : 임계값을 0.5로 설정

CatBoost가 1로 예측



XGBoost, LightGBM
둘 다 0으로 예측해도





Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

Lv.6

Lv.7

Lv.8

Submission Ensemble

Threshold Control

[CASE 1] : 임계값을 0.5로 설정

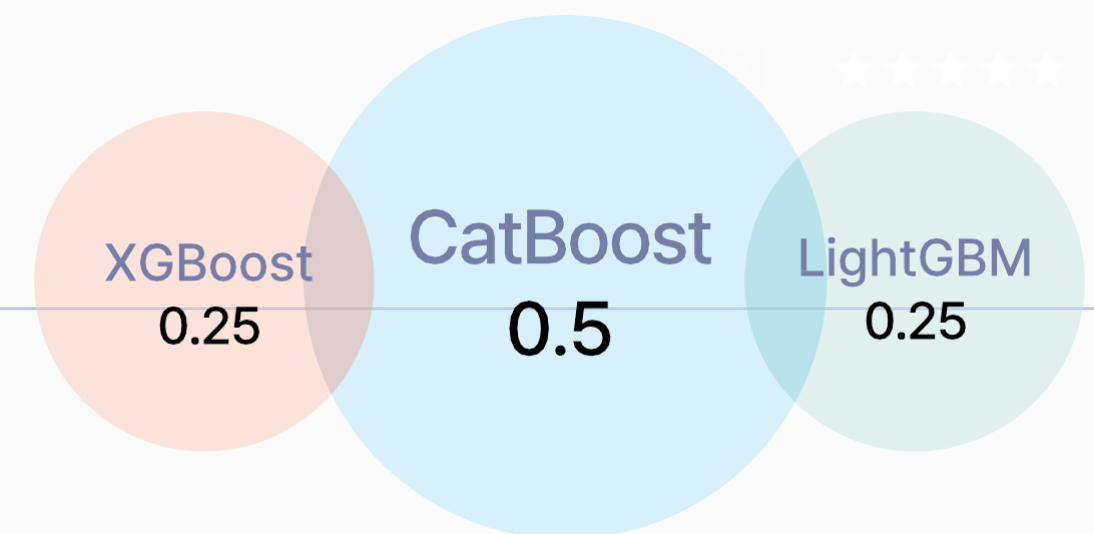
CatBoost가 1로 예측



XGBoost, LightGBM
둘 다 0으로 예측해도



최종 Label을 1로 예측





Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

Lv.6

Lv.7

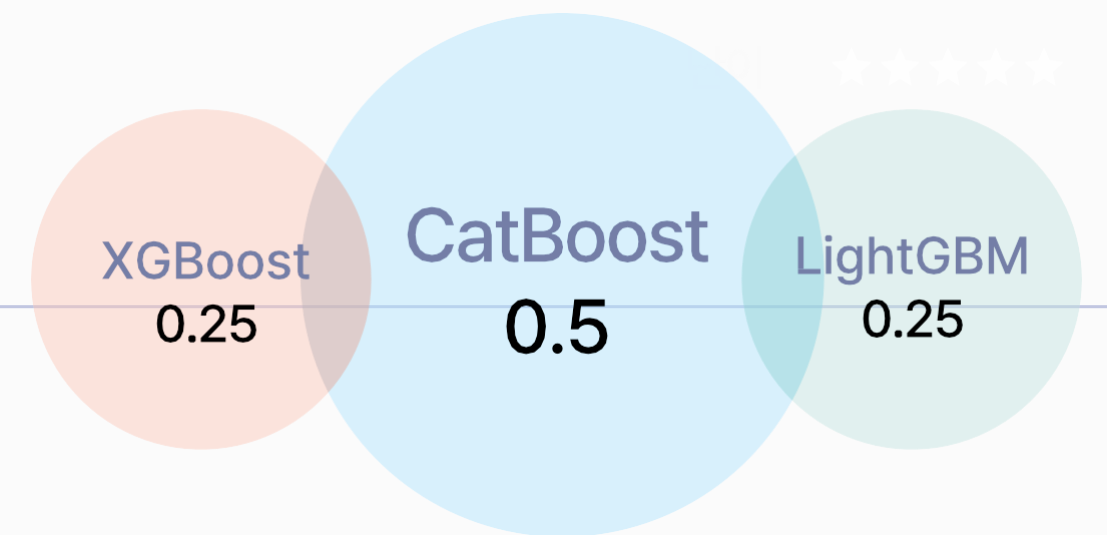
Lv.8

Submission Ensemble

Threshold Control

[CASE 2] : 임계값을 **0.5**로 설정

CatBoost가 **0**으로 예측





Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

Lv.6

Lv.7

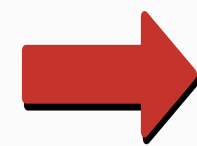
Lv.8

Submission Ensemble

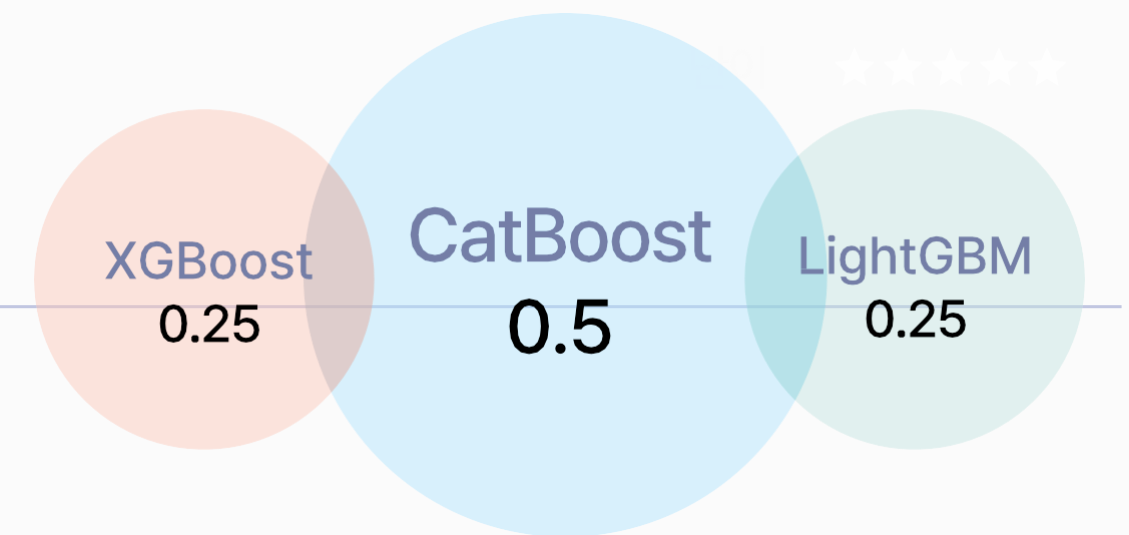
Threshold Control

[CASE 2] : 임계값을 **0.5**로 설정

CatBoost가 **0**으로 예측



XGBoost, LightGBM
둘 다 **1**로 예측해야만





Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

Lv.6

Lv.7

Lv.8

Submission Ensemble

Threshold Control

[CASE 2] : 임계값을 **0.5**로 설정

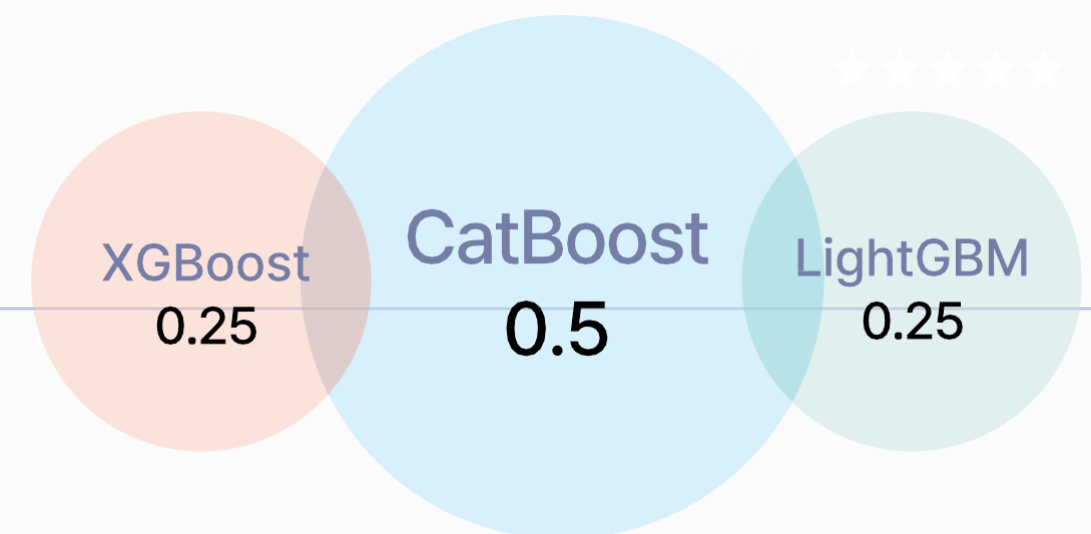
CatBoost가 **0**으로 예측



XGBoost, LightGBM
둘 다 **1**로 예측해야만



최종 Label을 **1**로 예측





Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

Lv.6

Lv.7

Lv.8

Public & Private Score



Public Score

0.67435

Private Score

0.64516



Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

Lv.6

Lv.7

Lv.8

Opinion & Feedback



[2분반] 3조

0.64516





Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

Lv.6

Lv.7

Lv.8

Opinion & Feedback



[2분반] 3조

0.64516



단계적 분석
부족



Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

Lv.6

Lv.7

Lv.8

Opinion & Feedback



[2분반] 3조

0.64516



단계적 분석
부족

과적합



Lv.1

Lv.2

Lv.3

Lv.4

Lv.5

Lv.6

Lv.7

Lv.8

Opinion & Feedback



[2분반] 3조

0.64516



단계적 분석
부족

과적합

모델링
치중 경향

★★★★★

THANK YOU